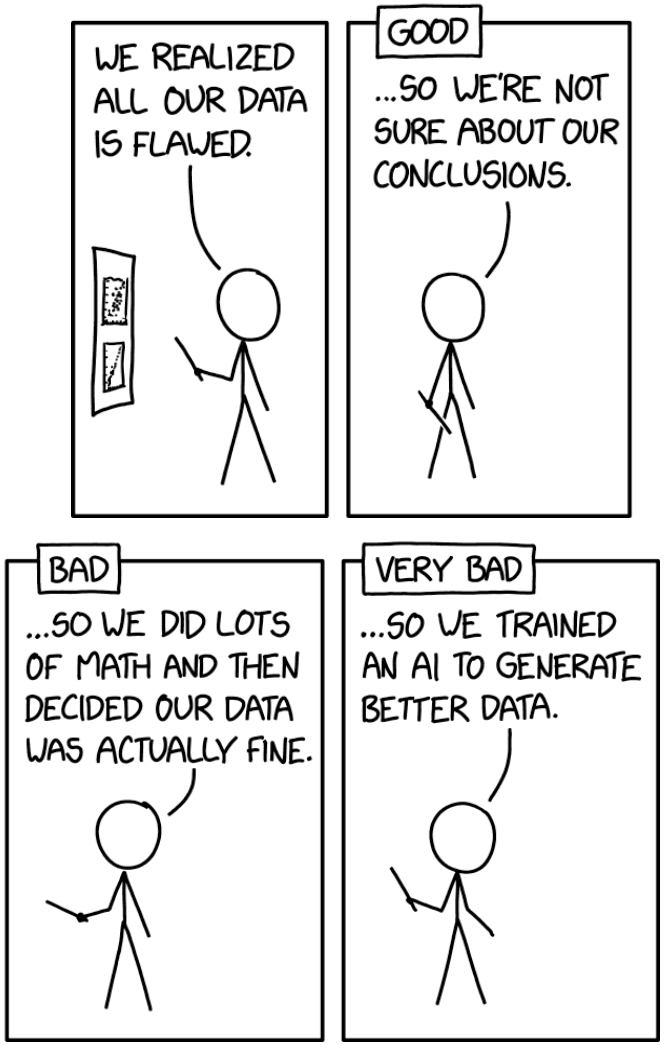
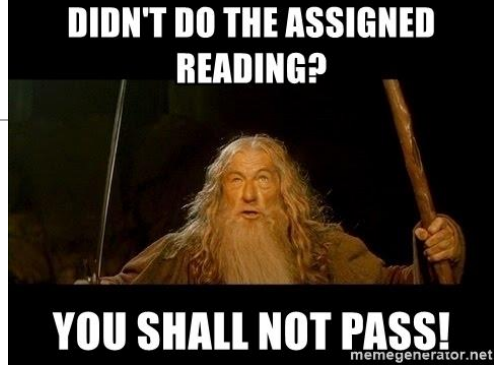
P-values and Bad Statistics

John Guttag

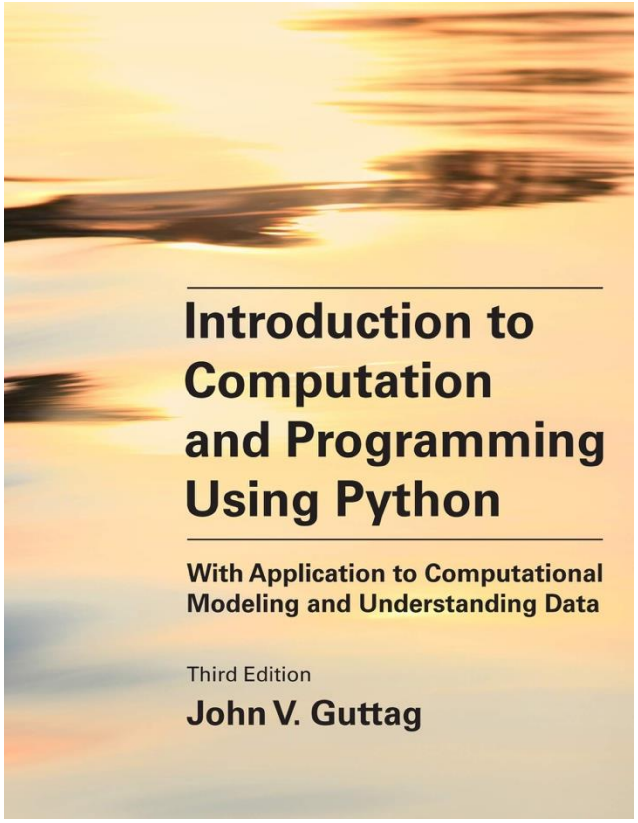
MIT Department of Electrical Engineering and Computer Science



Reading



- Today:
  - Sections 21.1 – 21.5
  - Chapter 22



Been Talking a Lot About Statistics

- Statistics is about finding ways to analyze data that allow us to draw conclusions despite uncertainty
- Allows us to derive conclusions about the world from limited samples
- Descriptive statistics
  - Describing the sample—these are factual statements
    - Plots, mean, median, variance, etc.
- Inferential statistics
  - Extrapolate from samples to assertions about the populations from which the samples are drawn
  - No guarantee that these are correct

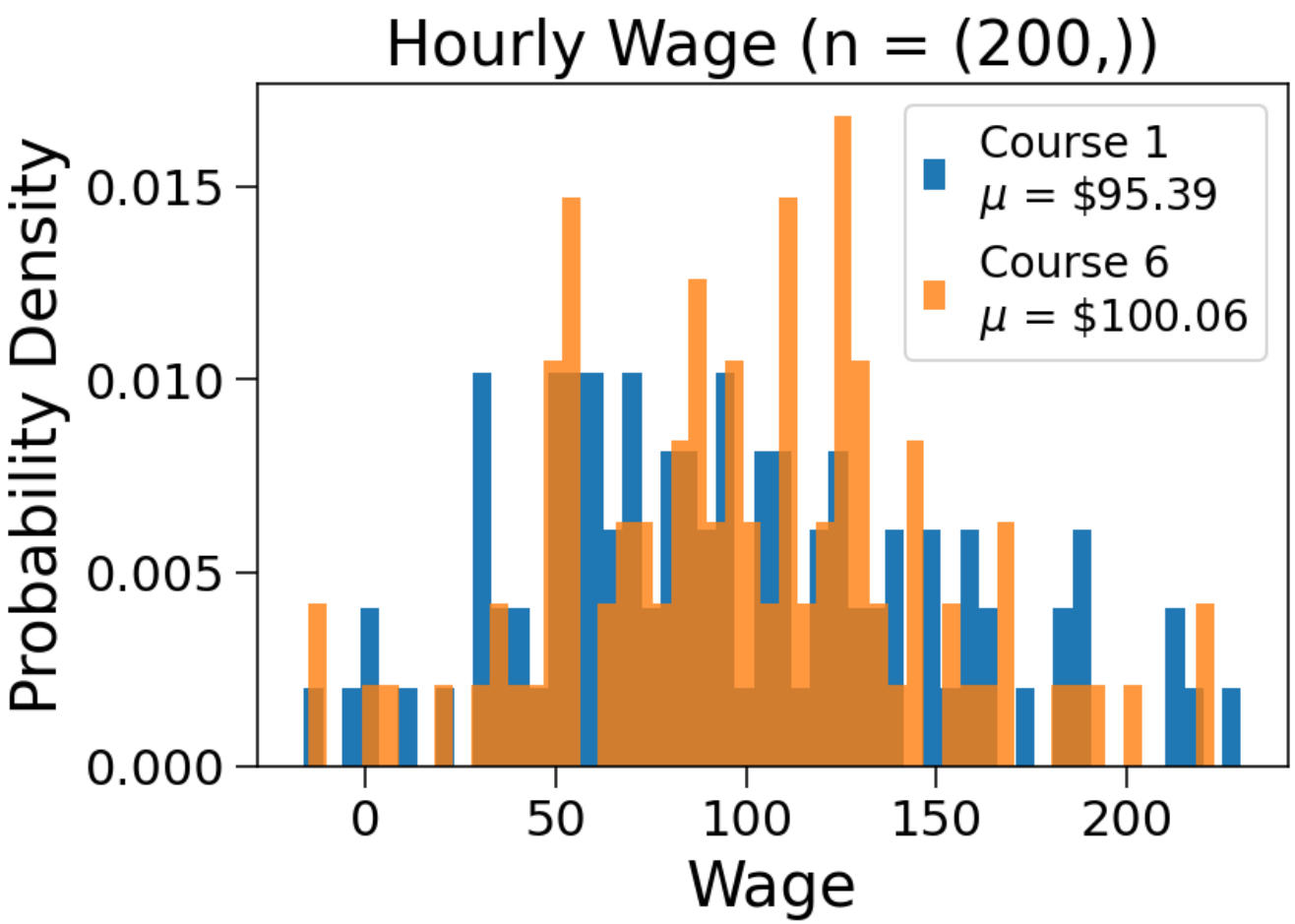
"All models are wrong, but some are useful."

--George Box

Trusting Inferences

- There are methods to quantify the uncertainty of an inference
- Already looked at putting confidence intervals around estimates
- Today, we will talk (briefly) about hypothesis testing
- Consider, for example, the hypothesis that Course 6 and Course 1 graduates have, on average, different hourly wages

A (made up) Sample of Hourly Wages from Each Population



Are the means of these two samples statistically significantly different?

Hypothesis Testing

- **Alternative hypothesis** (H_a)
  - Something that you think is implied by the data
  - E.g., Course 6 and Course 1 graduates have, on average, different hourly wages
- **Null hypothesis** (H_0)
  - The data does not support the alternative hypothesis
  - Any difference in the observed means might be attributed to random variation

Two Broad Classes of Hypothesis About Means

- The means are different
  - $\mathrm{H}_0 : \mu 1 \neq \mu 2$
- One mean is larger than the other
  - $\mathrm{H}_0 : \mu 1 \leq \mu 2$
- Should choose before looking at data

Testing a Hypothesis

- State H_a and H_0 unambiguously
- Compute a test statistic that takes into account
  - Magnitude of effect
  - Size of samples
  - Variance of samples

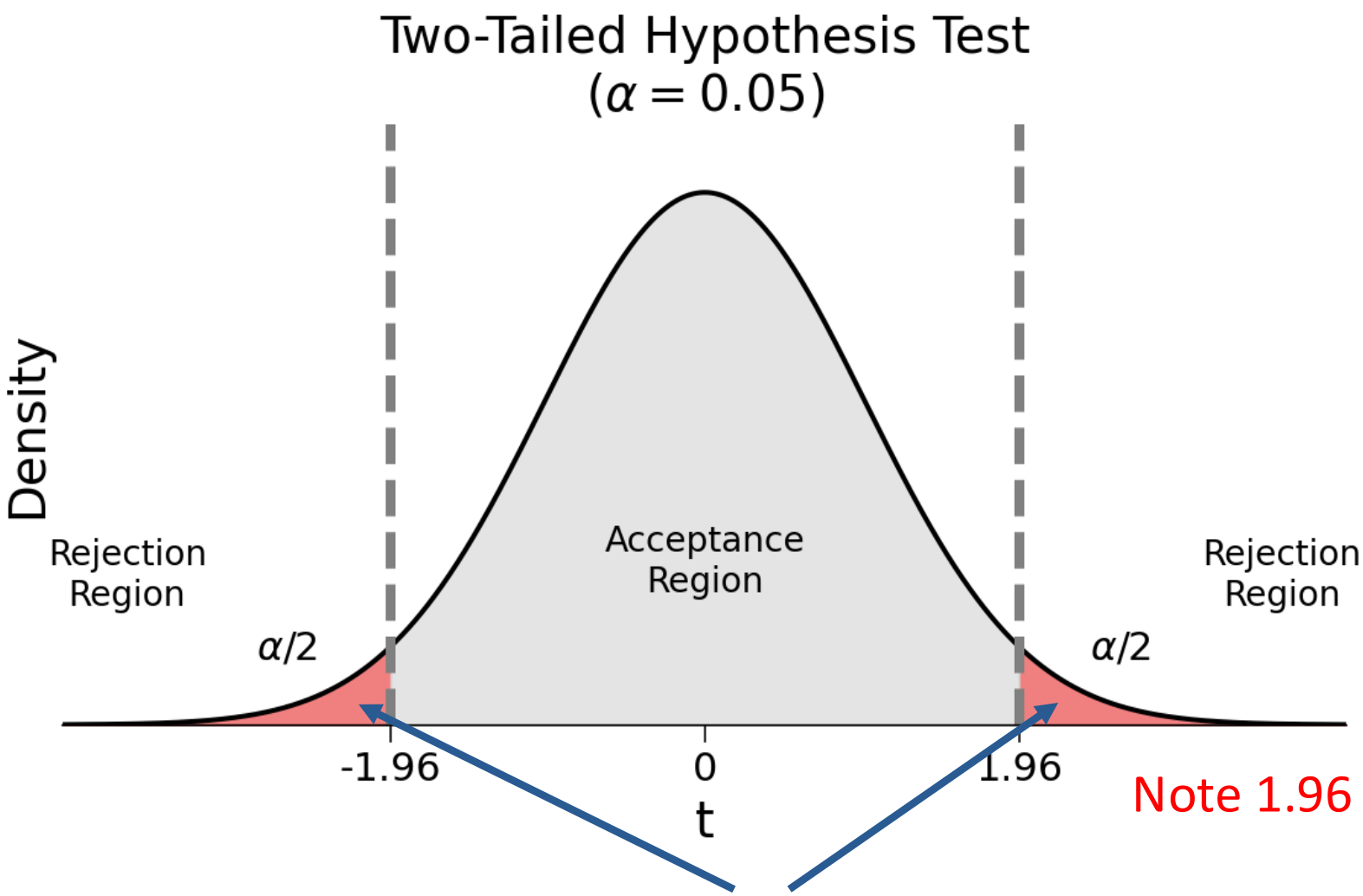
$$\frac{\mu 1 - \mu 2}{\sigma p \sqrt{\frac{1}{n1} + \frac{1}{n2}}}$$

Assumes variance of two populations is same

What does this remind you of?

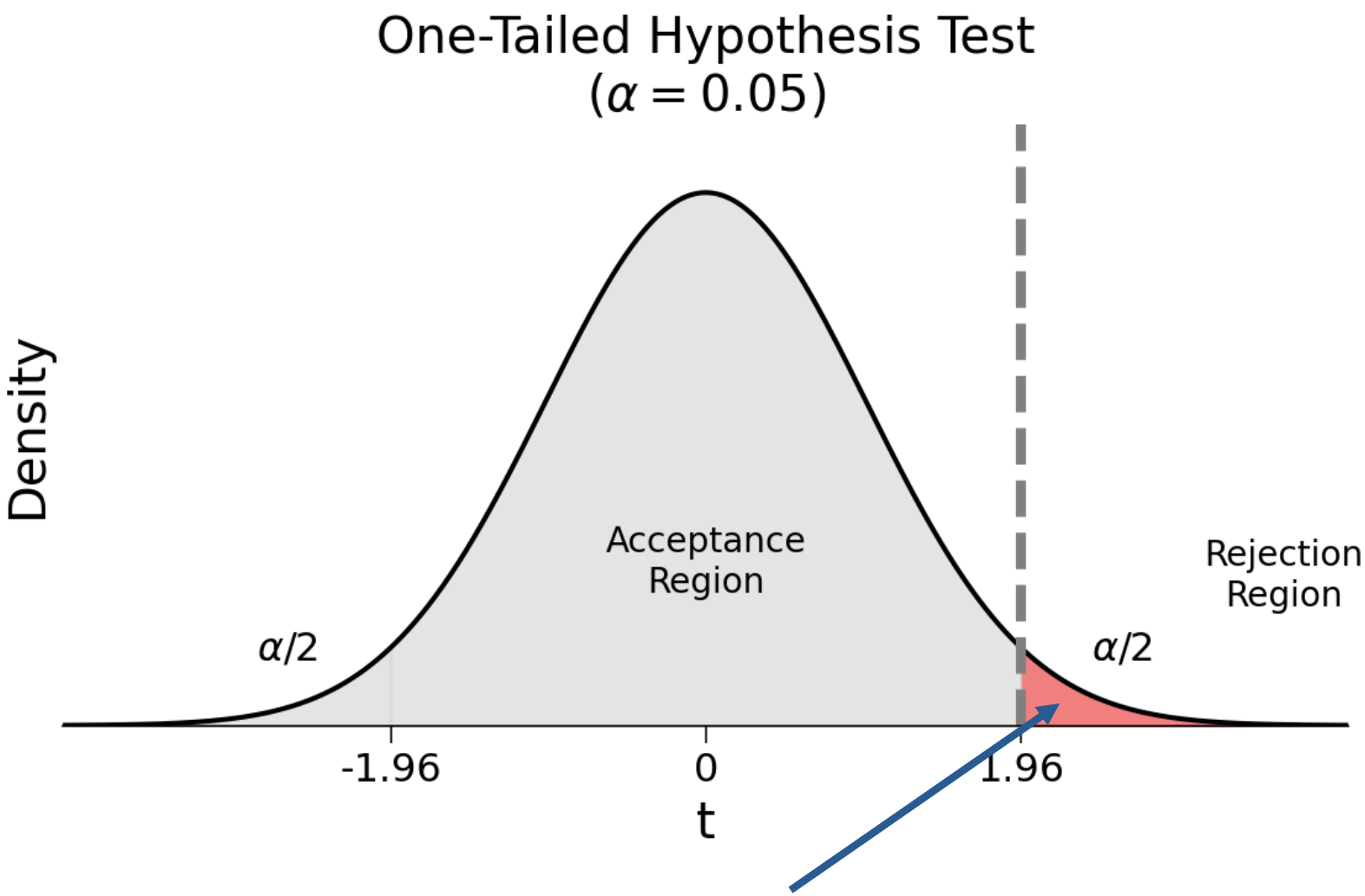
- Ask, if H_0 were true, what is the probability of seeing a test statistic at least as extreme as in the data?
  - This is called a **p-value**
- The p-value is a measure of how surprising the test test statistic would be **if** H_0 were true

A Pictorial Representation of a Two-tailed test ($H_0 : \mu 1 \neq \mu 2$)



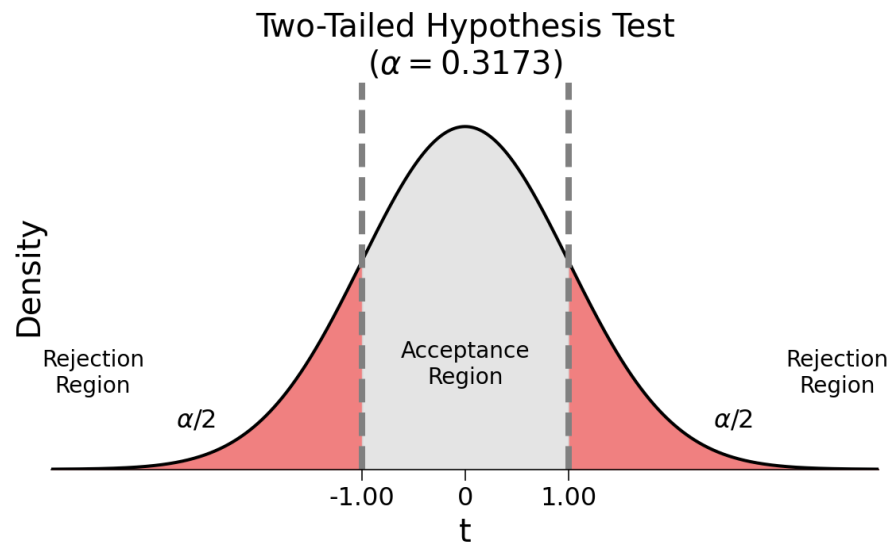
Area where we reject null hypothesis with 95% confidence

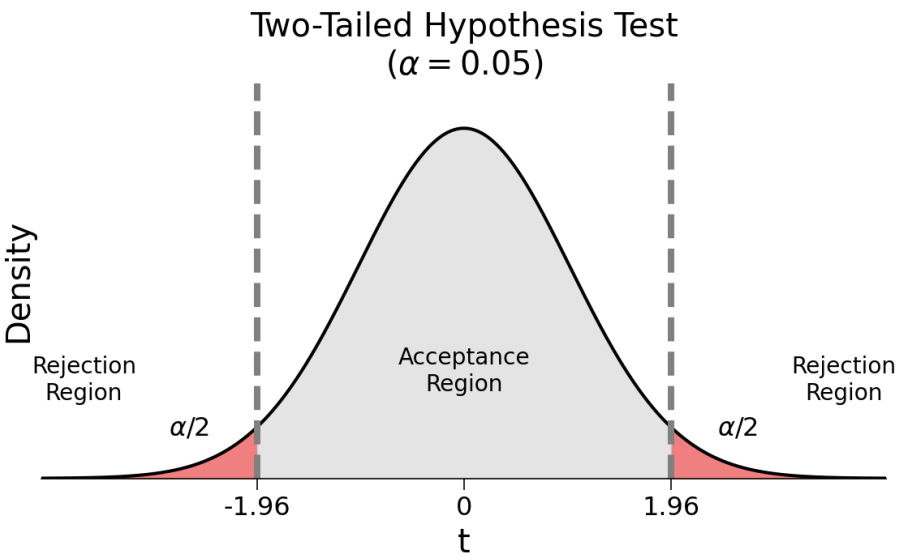
A Pictorial Representation of a One-tailed test ($H_0 : \mu 1 \leq \mu 2$)

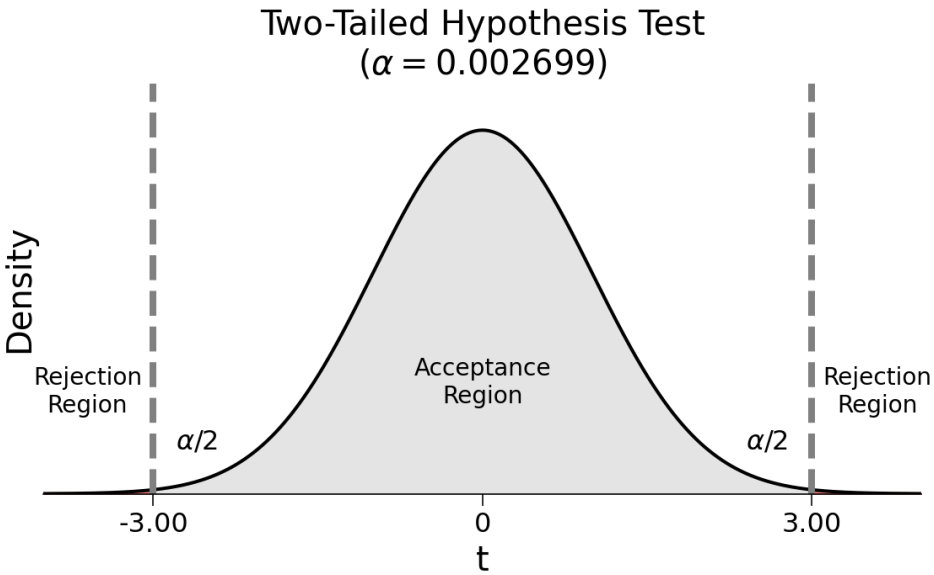


Area where we reject null hypothesis with 95% confidence

Empirical Rule: 1, 1.96, and 3 Standard Deviations







Checking the Hypothesis

```
import scipy.stats as stats
```

```
def compare_distributions(d1, d2, verbose = True):
    mean_1, mean_2 = sum(d1)/len(d1), sum(d2)/len(d2)
    t_val, p_val = stats.ttest_ind(d1, d2)
    if verbose:
        label_1 = f'Course 1\n$\mu$ = \${sum(d1)/len(d1):.2f}'
        label_2 = f'Course 6\n$\mu$ = \${sum(d2)/len(d2):.2f}'
        plt.hist(d1, bins = 50, density = True, label = label_1)
        plt.hist(d2, bins = 50, density = True, label = label_2,
                 alpha = 0.8)
        plt.legend(loc = 'best')
        plt.ylabel('Probability Density')
        plt.xlabel('Wage')
        plt.title(f'Hourly Wage (n = {(len(d1) + len(d2)):,})')
        print(f'Difference in means = {abs(mean_1 - mean_2):.4f}, '
              f'test-stat = {t_val:.3f}, p-value = {p_val:.3f}')
    return p_val
```

T-test

```
t_val, p_val = stats.ttest_ind(d1, d2)
```

Performs a **two-sample two-tailed t-test for independent samples** —it compares the means of two independent groups to see if they differ *significantly*. (Would use a 1-tailed t-test if H_a specified direction of difference.)

Returns a tuple of the t-statistic for the two samples and the probability, under the null hypothesis, of observing a t-statistic at least this far from zero.

What This Means

Difference in means = 4.67, $\boxed{\text{p-value} = 0.515}$

If there were actually no difference in the means of the two populations (i.e., if the null hypothesis were true), the probability of observing a difference of at least 4.67 in a random sample of 200 observations (100 per group) is approximately .515

It does not imply that the probability of the null hypothesis being true is .515

Analogous to “not guilty” vs. “innocent”

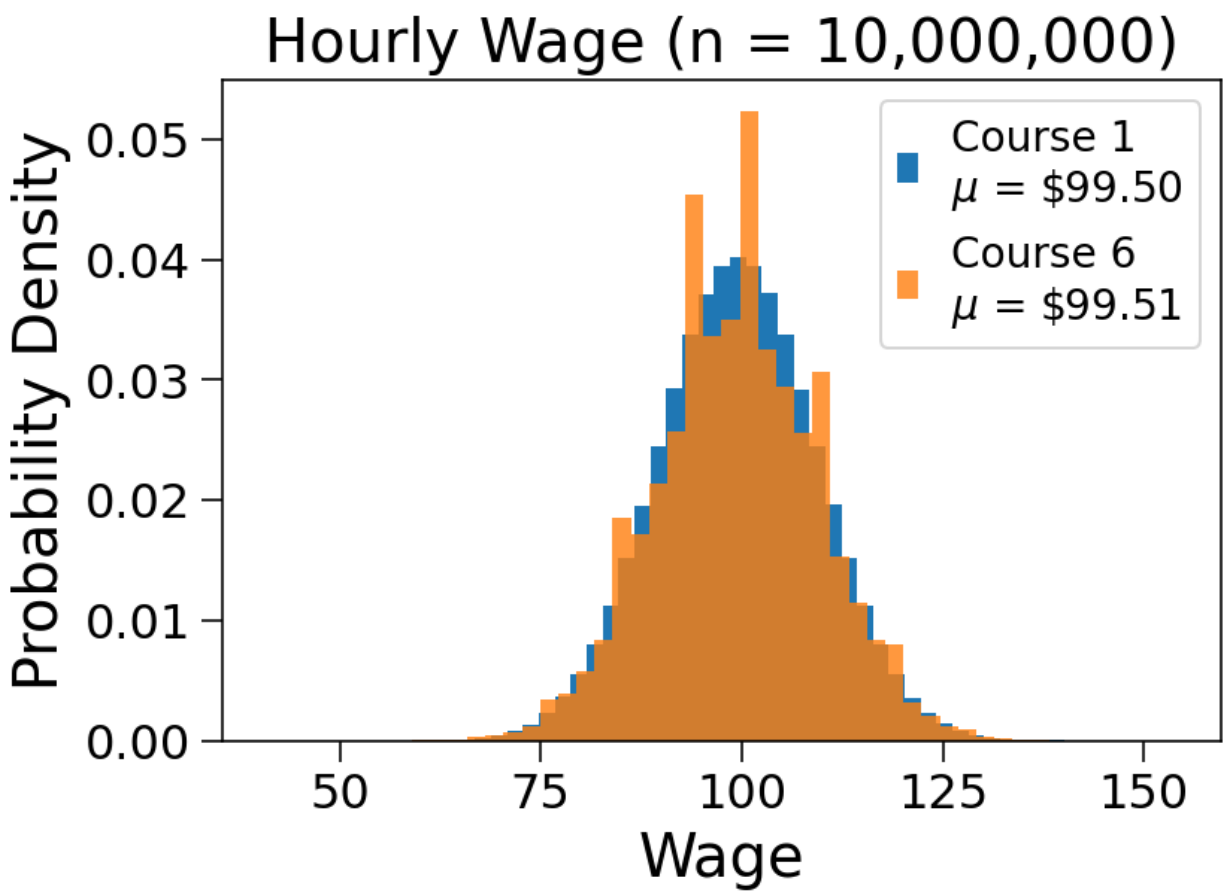
Aside: Suppose Alternative Hypothesis Were $\mu 1 > \mu 2$

```
def one_tail(d1, d2): # u1 > u2
    t_val, p_val = stats.ttest_ind(d1, d2)
    if t_val < 0:
        p_one_tailed = p_val/2
    else:
        p_one_tailed = 1 - (p_val/2)
    return p_one_tailed
```

```
print('Try mean(d1) > mean(d2)')
print(f'  test-stat = {one_tail(d1, d2)[0]:.3f}, '
      f'p_val = {one_tail(d1, d2)[1]:.3f}')
print('Try mean(d2) > mean(d1)')
print(f'  test-stat = {one_tail(d2, d1)[0]:.3f}, '
      f'p_val = {one_tail(d2, d1)[1]:.3f}')
```

Try mean(d1) > mean(d2)
  test-stat = -0.653, p_val = 0.257
Try mean(d2) > mean(d1)
  test-stat = 0.653, p_val = 0.743

When n is Large, Small Differences Can Become "Significant"



Difference in means = 0.0138, test-stat = -2.183, p-value = 0.029

Beware: Statistically Significant != “significant”

Would you change majors for a penny an hour?

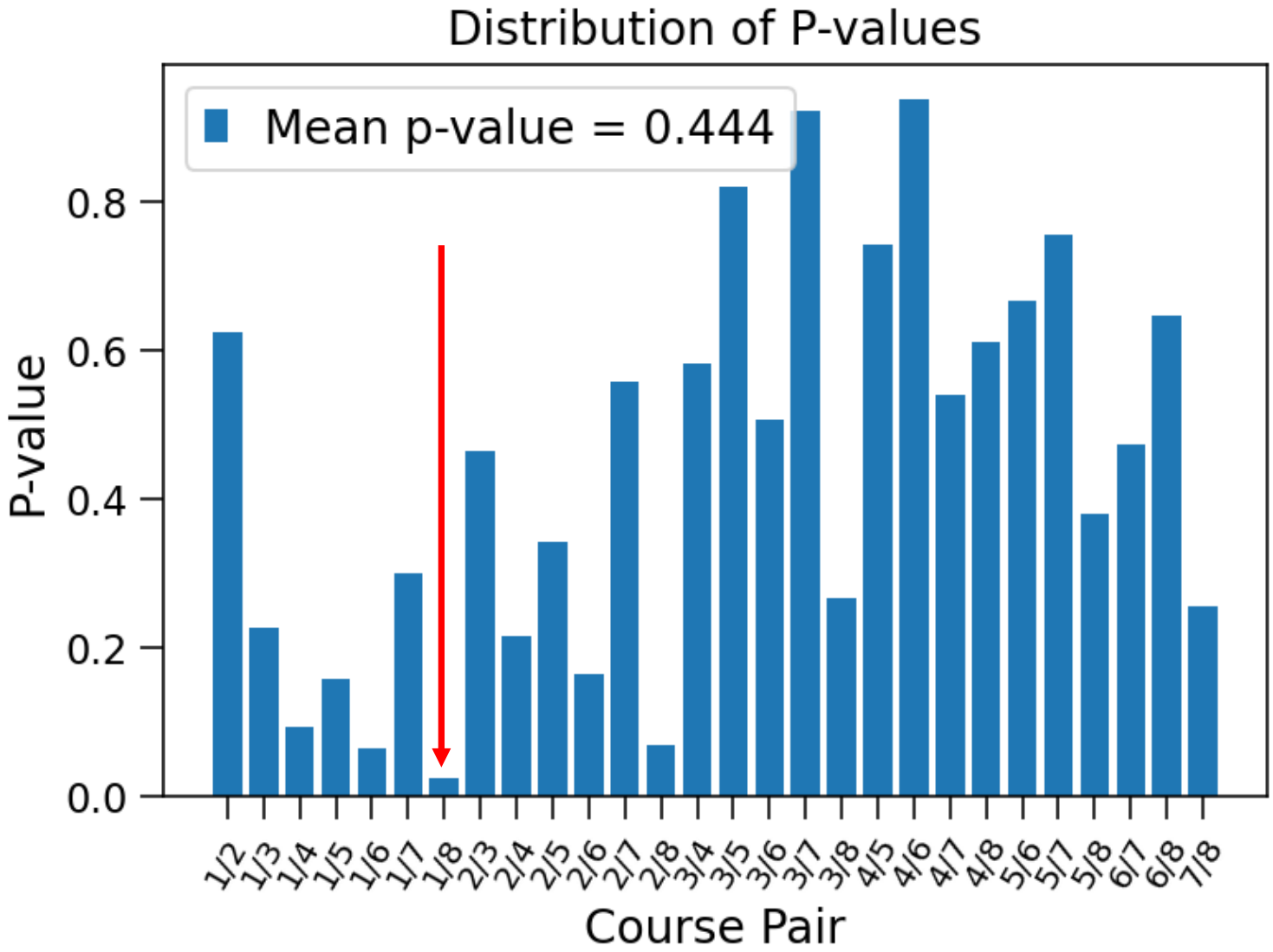
Let's Try Something Slightly Different

- We have samples from eight different courses
- Let's see if difference between any pair is statistically significant, by looking at all pairwise p-values
- It reports

The difference between Courses 1 and 8 is $-16.537, with p = 0.024

- Should we take this at face value?

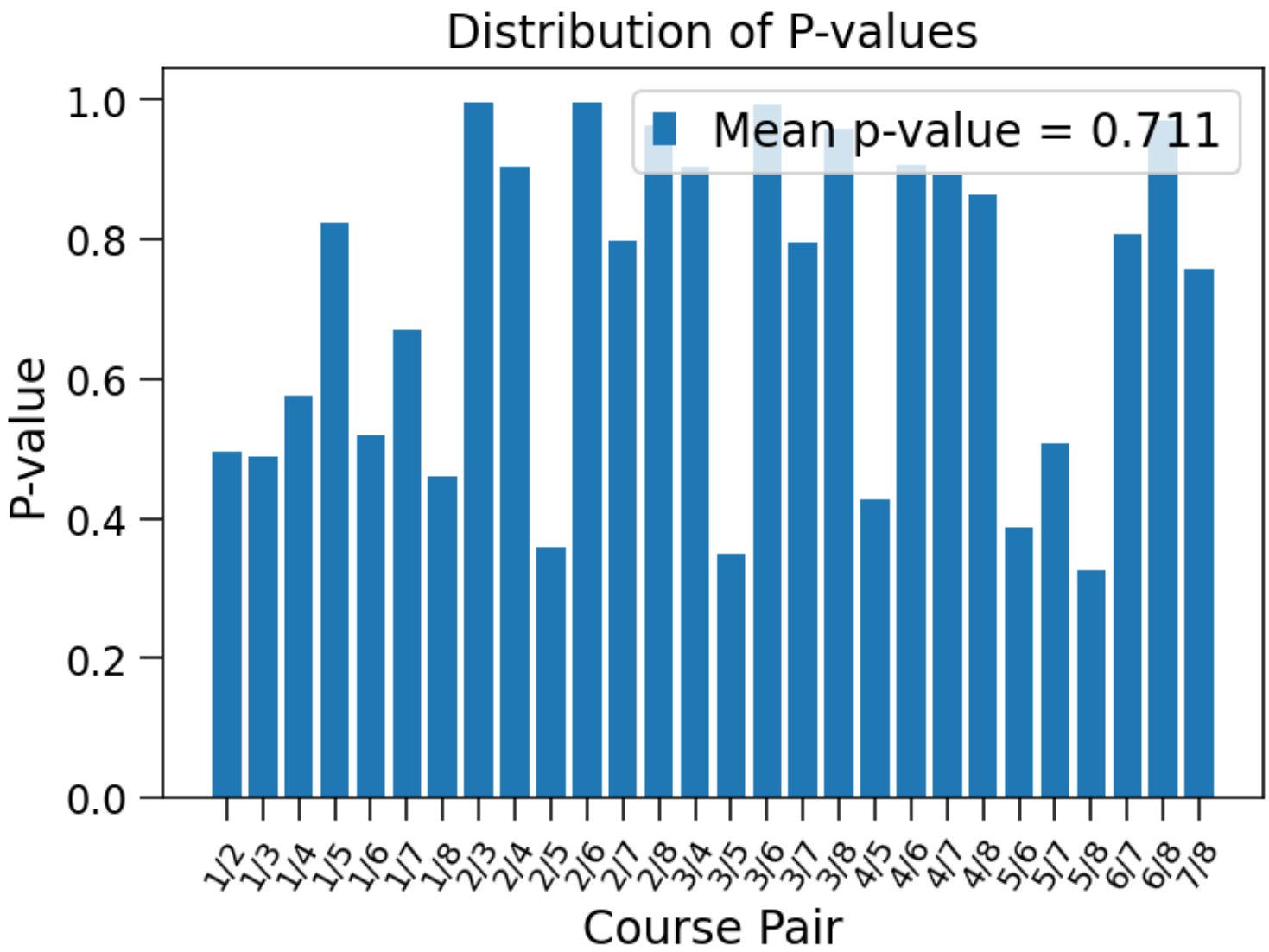
Distribution of P-Values



What Did We Do?

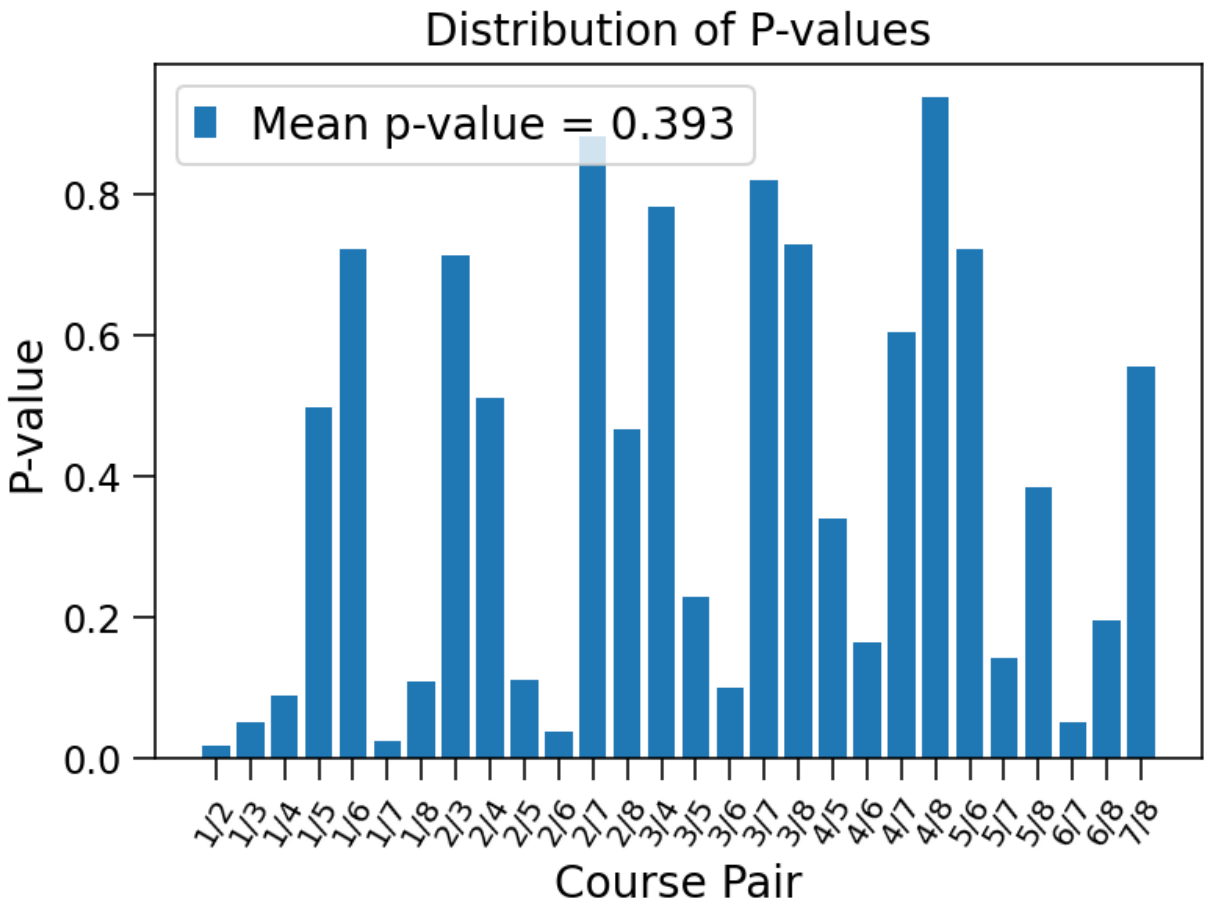
- Tested against 28 null hypotheses
- Suppose the probability of a false positive is 0.05 and each hypothesis is independent?
- Probability that none are accepted is
  - (1– 0.05)**28 = 0.24
- Probability that at least one false positive accepted is
  - 1 - 0.24 = 0.76
- Let's try changing seed (and nothing else) and running experiment again

Distribution of P-Values (different seed)



No significant differences

Distribution of P-Values (yet a different seed)



The difference between Courses 1 and 2 is $17.110, with p = 0.017

The difference between Courses 1 and 7 is $16.053, with p = 0.024

The difference between Courses 2 and 6 is $-14.640, with p = 0.038

Bonferroni Correction

- If we are checking a family of *n* hypotheses and want a significance level for all of them of *a*, we should have a significance level of *a/n* for each
  - O.05/28 = 0.0018
  - Nothing in any of the pairs meets that level
- Good thing, because samples for each of the courses was drawn from the same normal distribution

Wikipedia

Carlo E. Bonferroni
1892- 1960

Bonferroni Correction

- When in doubt, use it
- Conservative
  - Assumes that test statistics for hypotheses are uncorrelated
  - Reduces probability of false positives
  - Increases probability of false negatives
- Not always obvious when it is needed
- That said, more often not used when it should be than used when inappropriate

Start of a Table from a Recent Article

Table 1.

Reviews investigating the prevalence of adjustments for multiple comparisons in clinical and epidemiological studies

Study (first author)	Year published	Study interval	Scientific field	Number of studies investigated	Number of studies with multiple comparisons	Proportion of studies with adjustments for multiple testing
Wason *et al.*	2014	2012	Multiarm clinical trials	59		51%
Tyler *et al.*	2011	January 2007 and October 2008	Neurology and psychiatry, clinical trials	55	nearly half (52 reported ≥2 primary or secondary outcomes)	3/52=5.8%
Vickerstaff *et al.*	2015	July 2011–June 2014	Neurology and psychiatry	209	60 (29% reported multiple primary outcomes)	15/60=25%
Kirkham *et al.*	2015	2012	Otolaryngology, original human subjects research with samples of >100 subjects	195	140	10%
Stacey *et al.*	2012	2010	Ophthalmology, abstracts presented at ARVO 2010	6415 abstracts	538 (with more than five hypotheses)	14%
Dworkin *et al.*	2016	January 2006 and June 2013	Pain, randomized clinical trial	101	29	21%

Which gets us to

Multiplicity corrections in life sciences: challenges and consequences

Otília Menyhárt, Balázs Győrffy

International Journal of Epidemiology, Volume 54, Issue 4, August 2025, dyaf098, https://doi.org/10.1093/ije/dyaf098

Published: 27 June 2025 **Article history**

There are Three Kinds of Lies

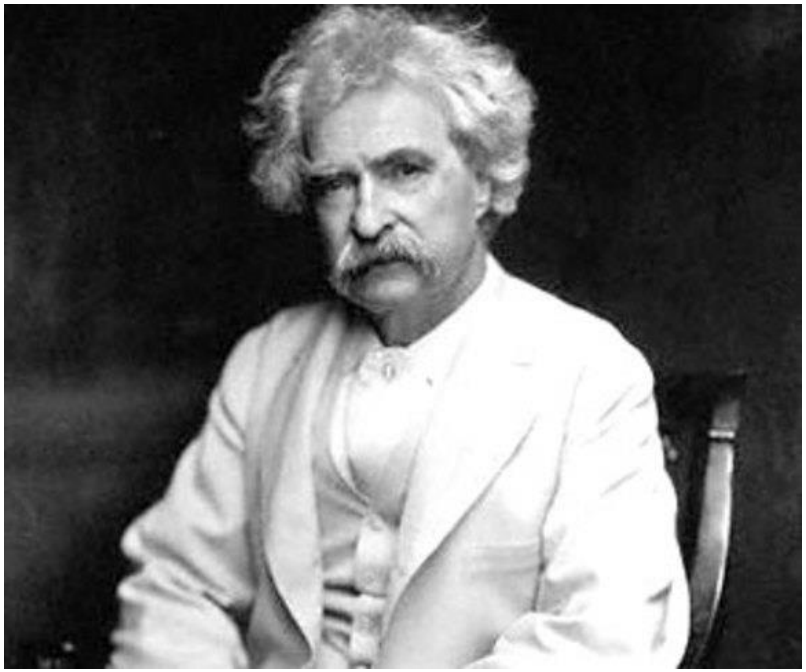
LIES

DAMNED LIES

and

STATISTICS

Attributed To



Using Numbers to Mislead

- Anyone can mislead by using inaccurate numbers
- "It used to be 1 in 20,000, then 1 in 10,000. And I would say that's probably 18 years ago. And now it's 1 in 31."

  Donald J. Trump, September 22, 2025
- "78.7% of MIT graduates say that taking 6.100B is better than sex."

  John Guttag, April 1, 2026

Today, I will help you learn how to mislead using accurate numbers

It's True, But …

- "less than 10 percent of documented transmission [of Covid-19], in many studies, have occurred outdoors"
  - Dr. Rochelle Walensky, CDC Director, April 27, 2021

Poll: What do you think actual fraction is?

10%
5%
2%
1%

It's True, But …

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  - Dr. Rochelle Walensky, CDC Director, April 27, 2021

Poll: What do you think actual fraction is?

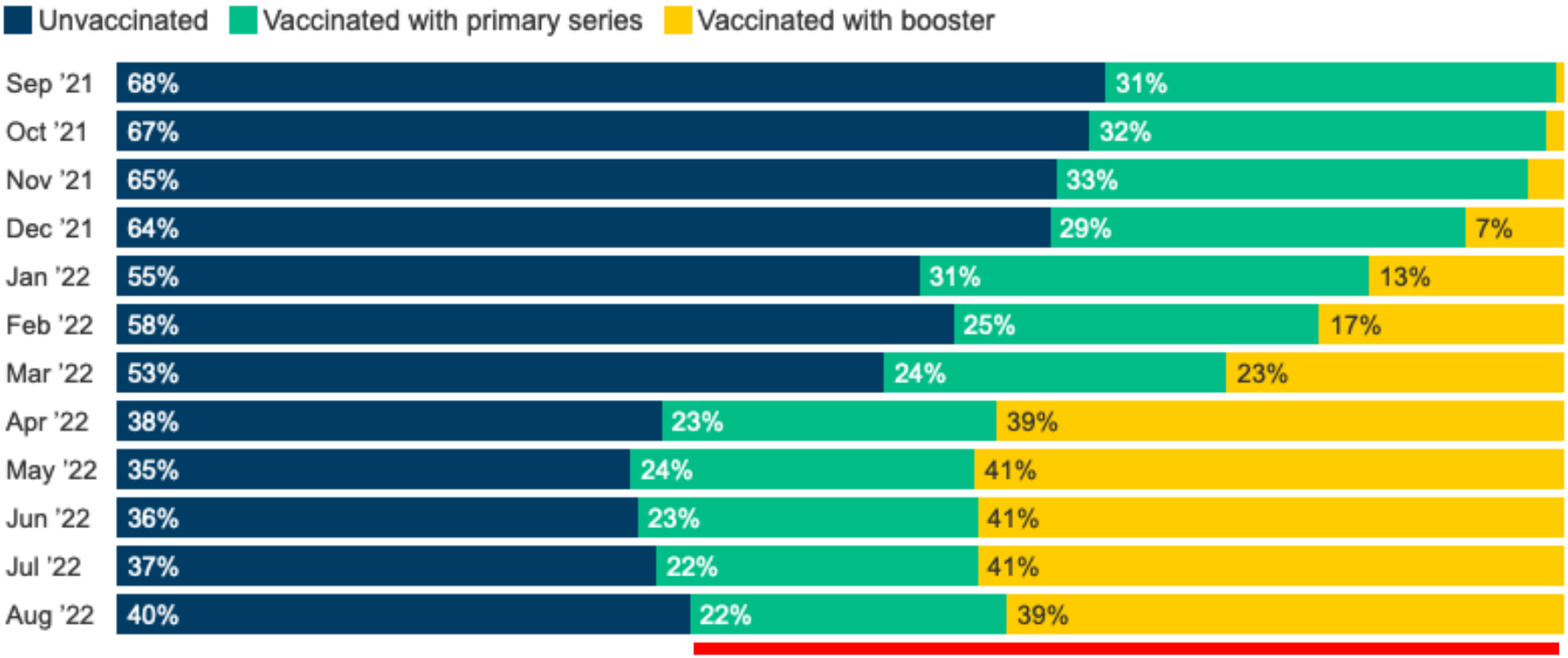
10%
5%
2%
1%

Estimate is <1%

Moral: Spurious baselines are useful for constructing statements that are both **true** and **deceiving**

Baseline Fallacy in Action, August 2022

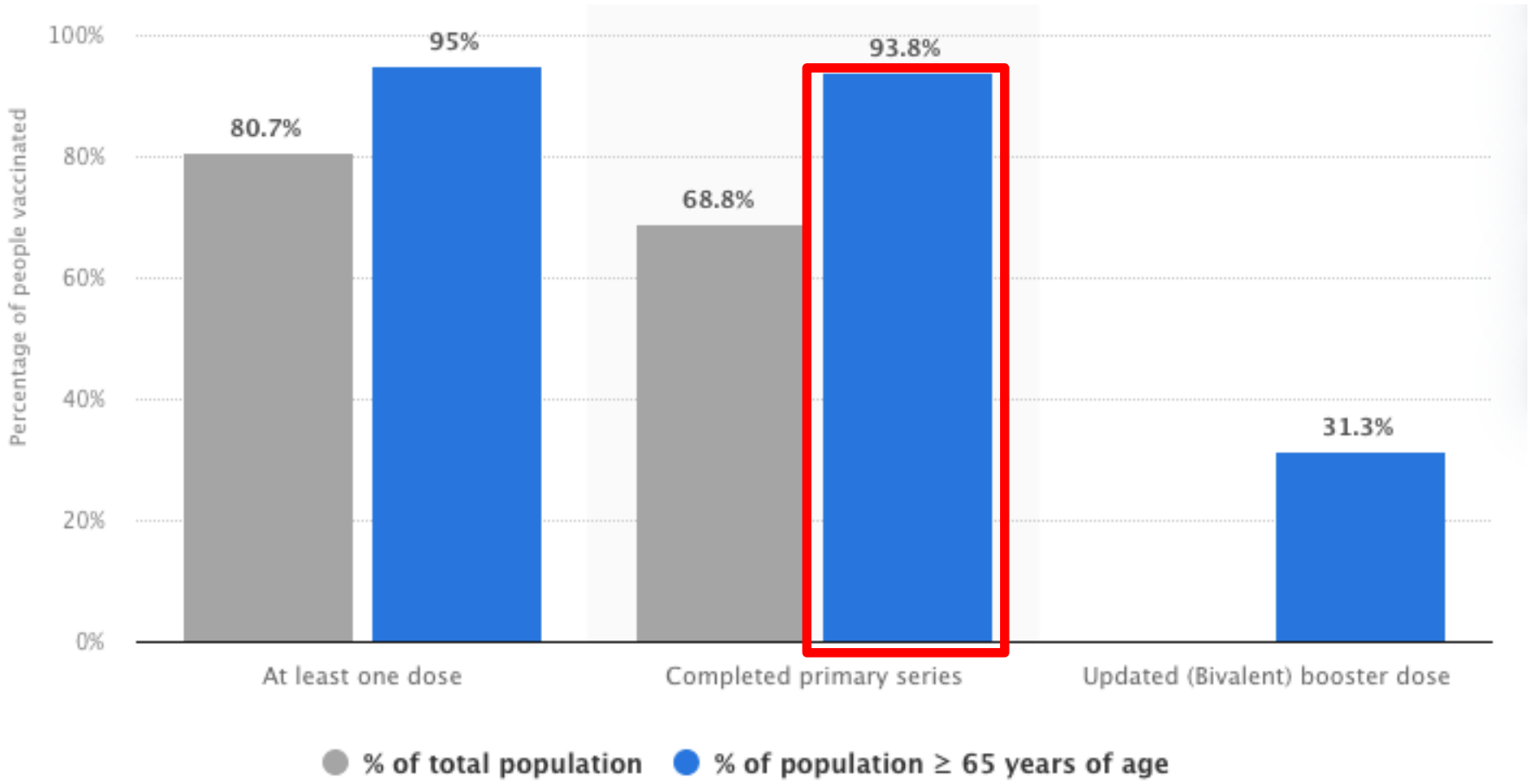
Most of the elderly dying from Covid have been vaccinated!



https://www.kff.org/covid-19/why-do-vaccinated-people-represent-most-covid-19-deaths-right-now/

Should the elderly return their vaccinations?

Maybe Not



40% of deaths were in the ~6% of population that were unvaccinated

Context Matters

The NEW ENGLAND JOURNAL of MEDICINE

- "researchers in Israel reported that vaccination increased the 42-day risk of myocarditis by a factor of 3.24 (95% confidence interval [CI], 1.55 to 12.44)" –*NEJM*, December 4, 2021
- Is this alarming?
- What do we need to know to answer that?

rate for unvaccinated = 0.002063%
X 3.24 = 0.006684%

Moral: Beware of percentage or multiplicative change when you don't know the baseline

p.s. People who contract Covid-19 are **16** times more likely to get myocarditis

Base-rate Fallacy

- Portable breathalyzer's have false negative rate of ~0% and a false positive rate of ~3%
  - I.e., 3% of tested sober individuals will be classified as DUI
- At checkpoints, police tested 1000 randomly chosen drivers. What is the probability that a driver who has a positive test is actually DUI, I.e., what is the PPV?
  - ~0.97?
- Assume that 0.5% of drivers are DUI
- For every 1000 drivers tested
  - 5 TP + 995*0.03 FP ≈ 35 positive tests
  - 5/35 ≈ 14% PPV

Moral: if you are drunk, DON'T DRIVE! If you are sober, don't take the test?

Moving On to Some Statistical Measures We Have Looked At

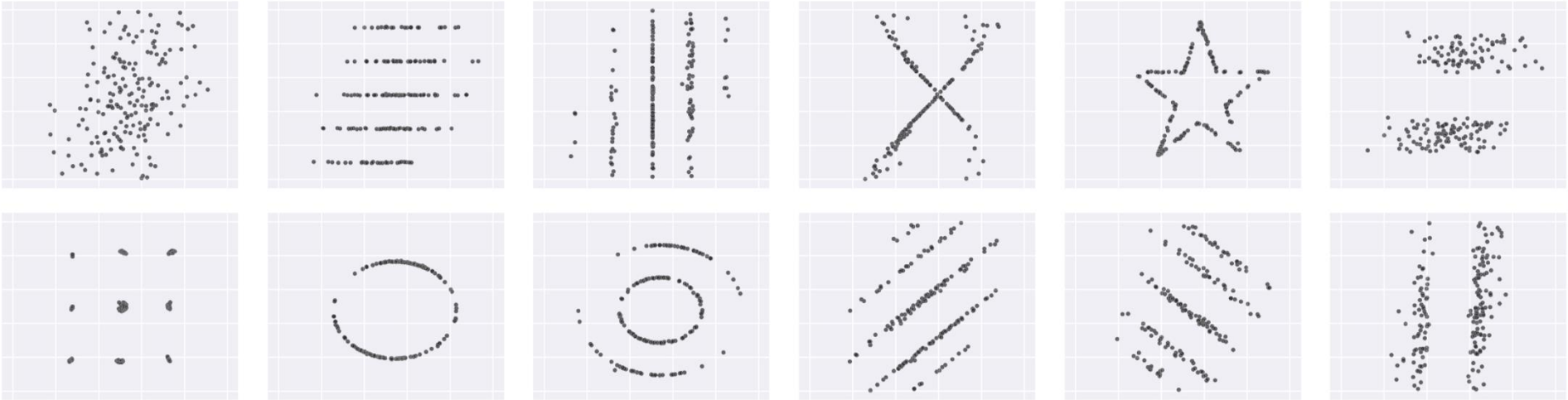
- https://en.wikipedia.org/wiki/Datasaurus_dozen is a well know public domain data set of 2D images.
- The following is true of each image in the dataset

```
X Mean: 54.26
Y Mean: 47.83
X SD  : 16.76
Y SD  : 26.93
Corr. : -0.06
```

Are these accurate statistics sufficient to understand the data ?

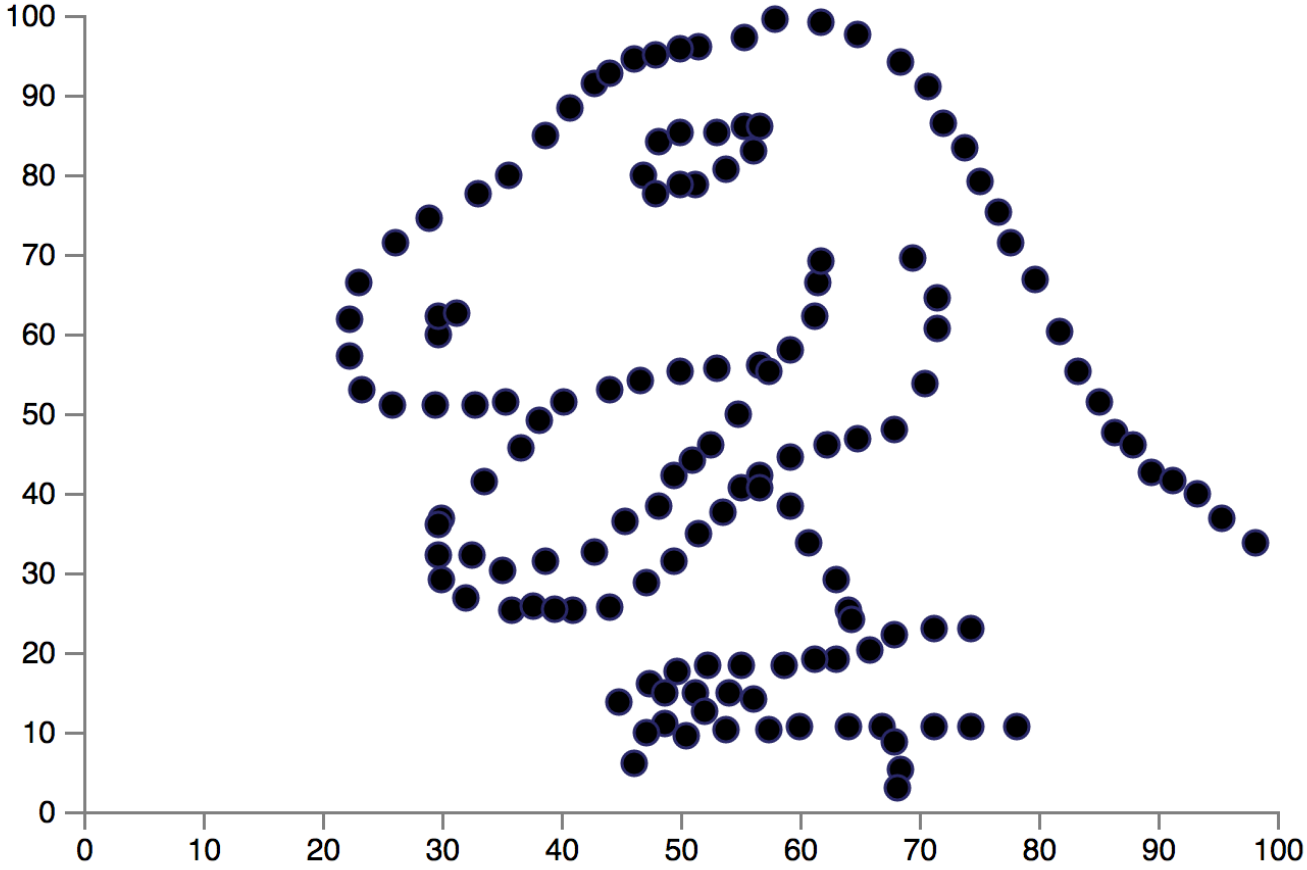
Not Really

```
X Mean:  54.26
Y Mean:  47.83
X SD   :  16.76
Y SD   :  26.93
Corr.  :  -0.06
```



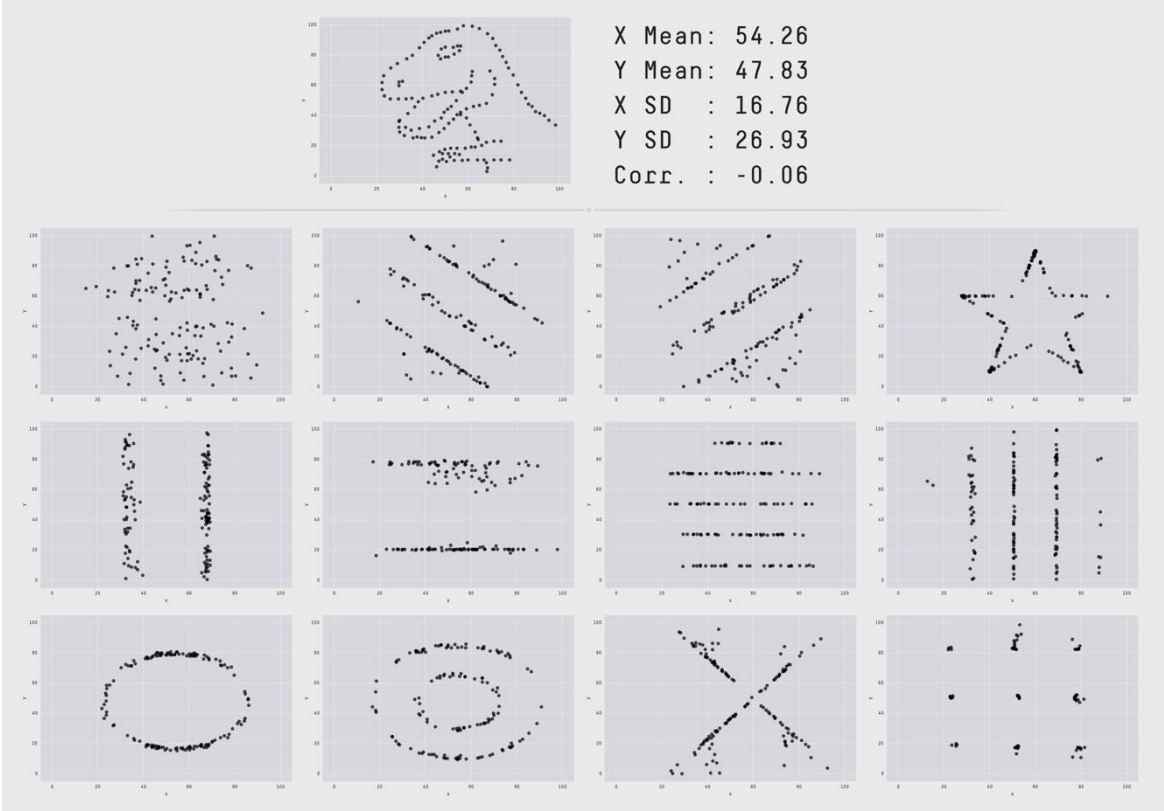
All have same means, standard deviations and correlations (up to two decimal places)

And This One Too



https://cran.r-project.org/web/packages/datasauRus/vignettes/Datasaurus.html

The Datasaurus Dozen



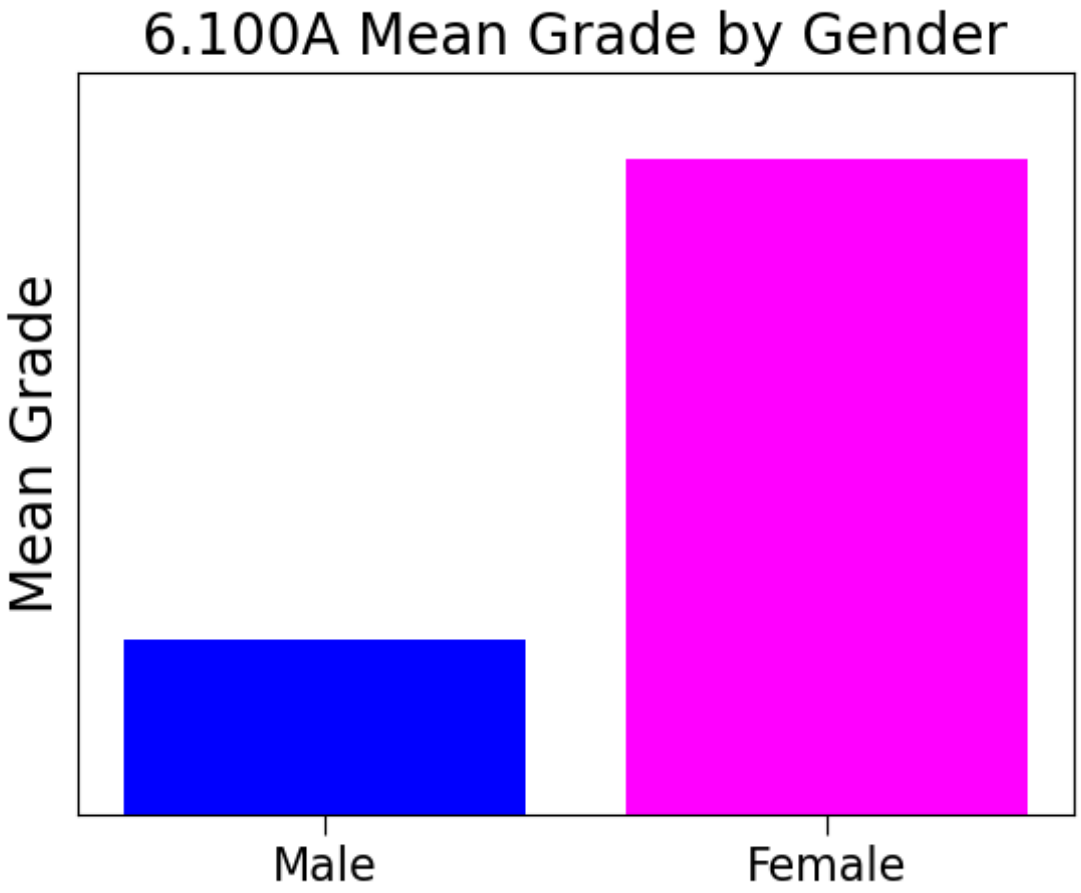
Moral: Statistics about the data are not the same as the data

Moral: Use visualization tools to look at the data itself

图片值1000字

Break Time

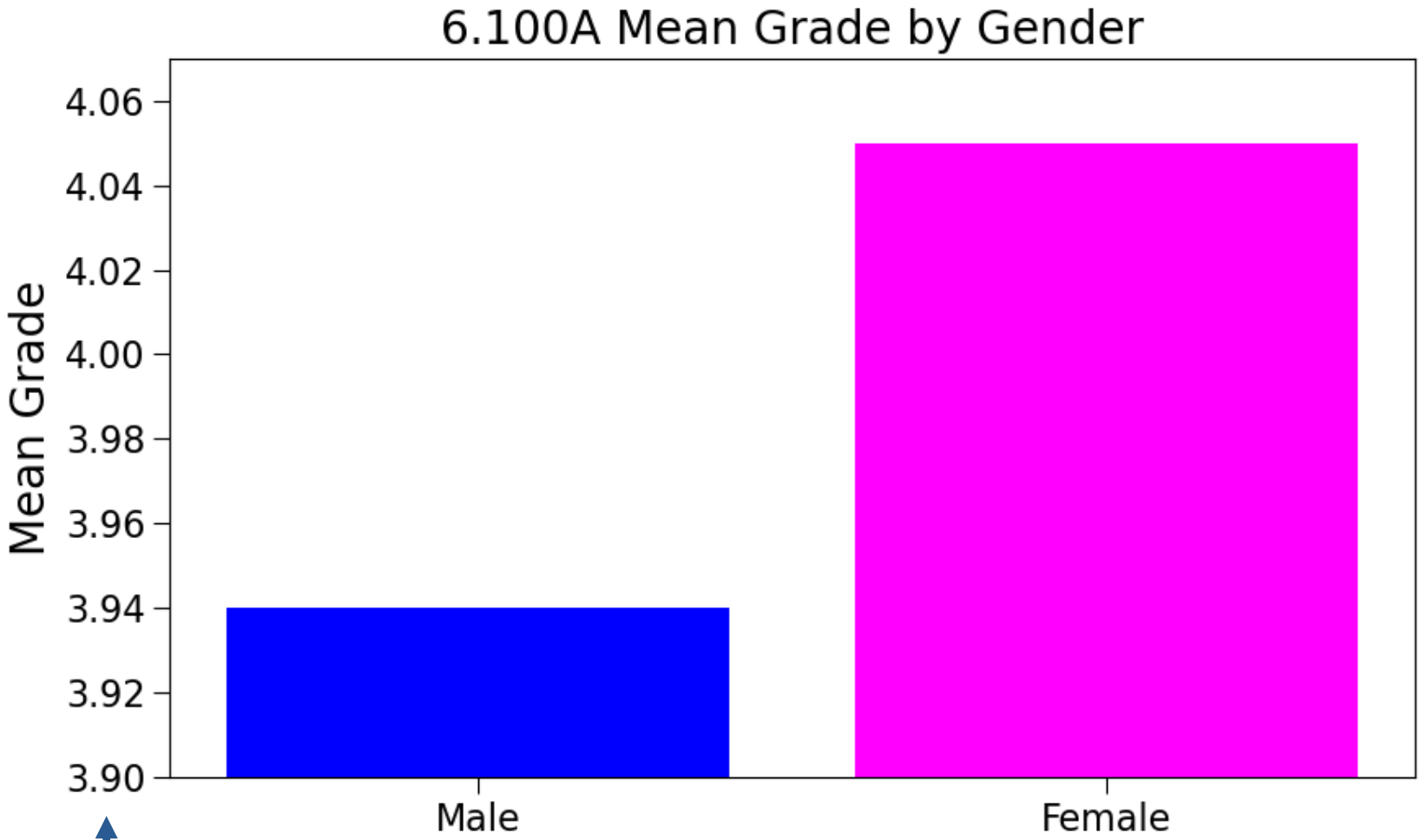
Is a Picture Always Worth 1000 Words?



This plot is best explained by

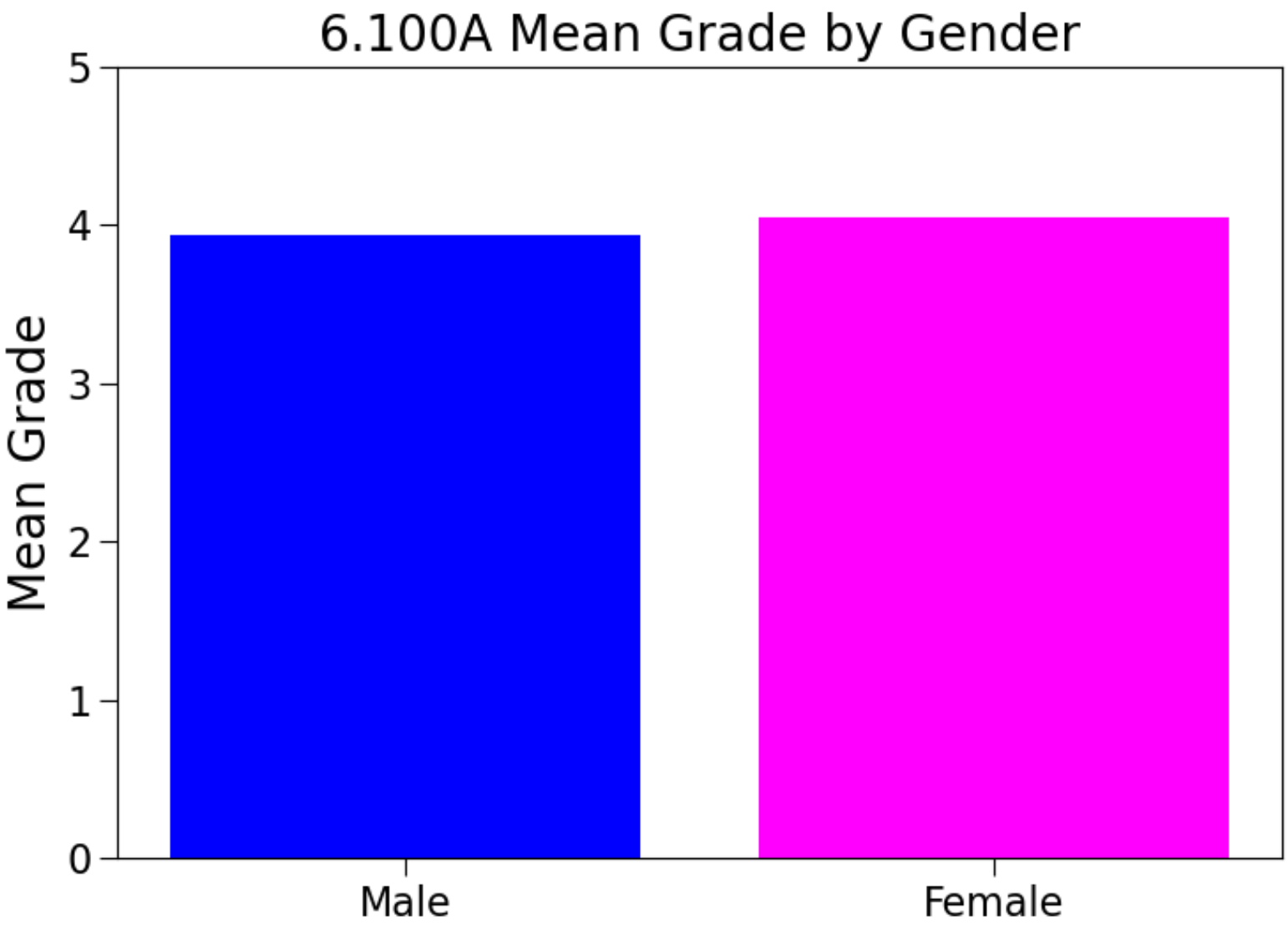
1. Women are better coders
2. The plot is fabricated
3. Bias in grade assignment
4. None of the above

Is a Picture Always Worth 1000 Words?



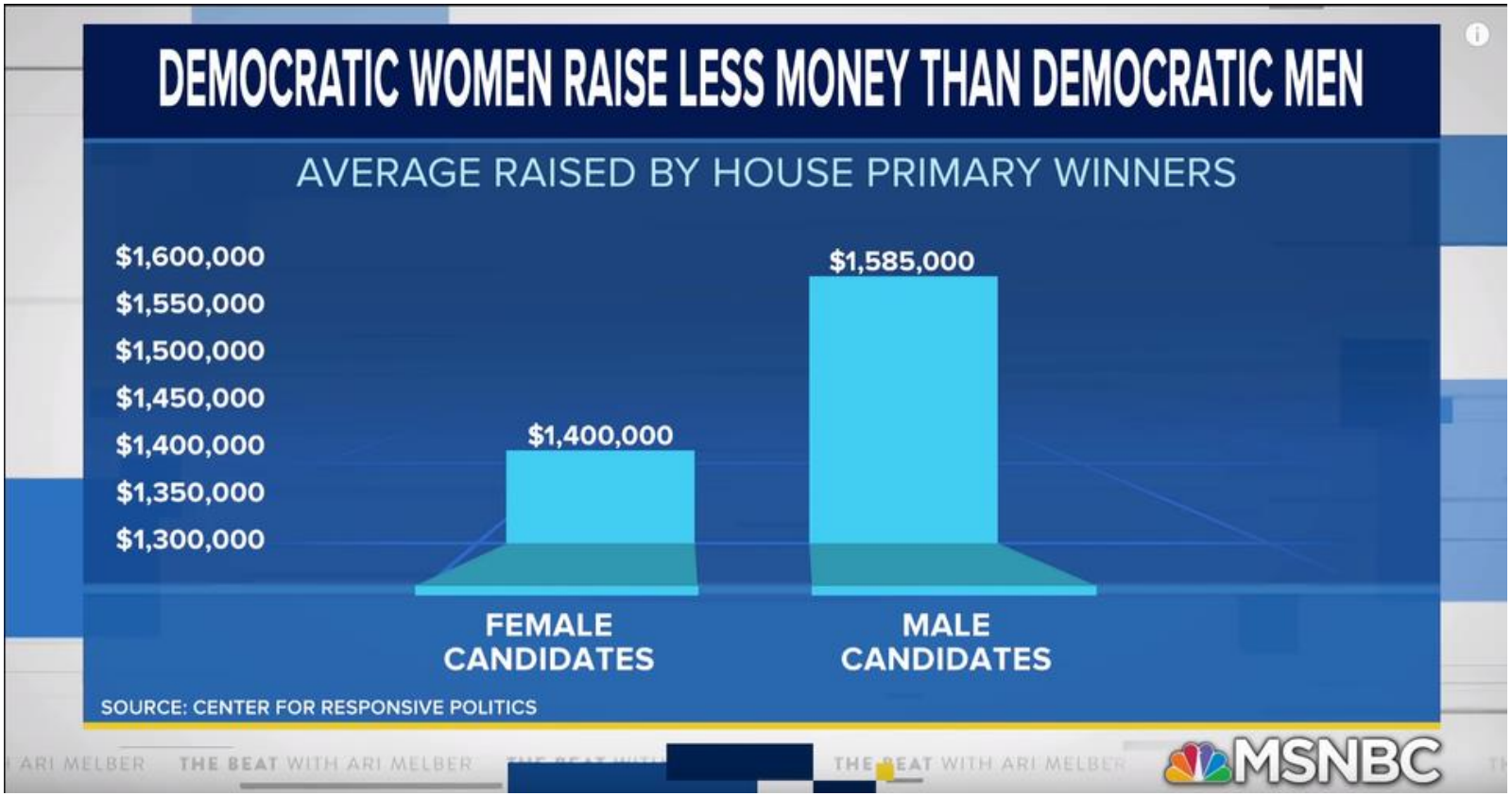
Moral: Beware of axis without values

The Same Data, Different Picture

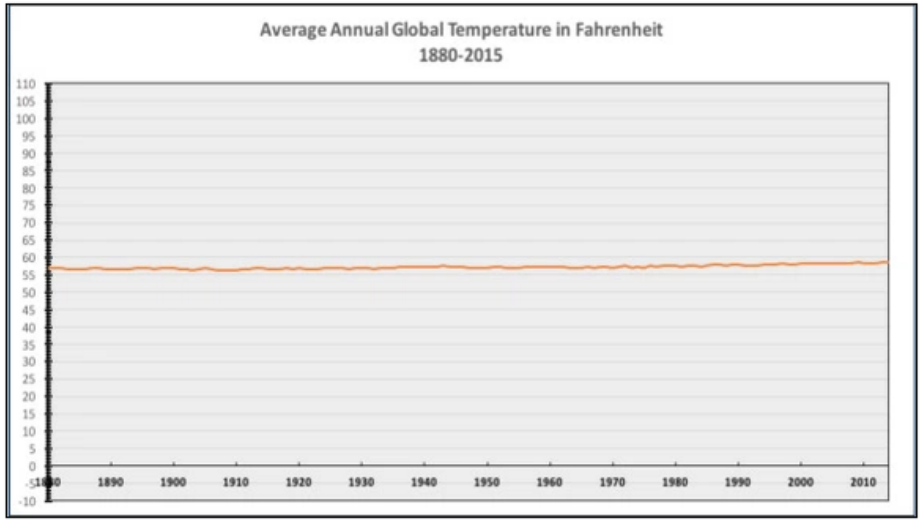


Moral: Beware of truncated y-axis

Example of Truncated Y-axis

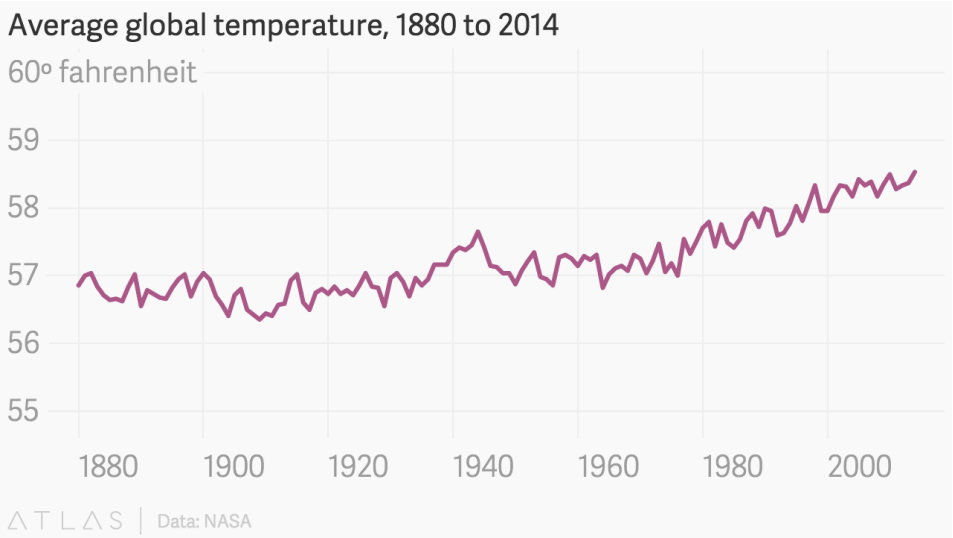


Should the y-Axis Always Start at 0?



Which conveys more information?

National Review, December 2015



Moral: OK to truncate the y-axis to remove preposterous values

Some Other Ways to Deceive with Plots

Is it linear?

Some Other Ways to Deceive with Plots

Spacing on x-axis
9, 6, 15 months

6 month period occupies more horizontal space than 15 month period

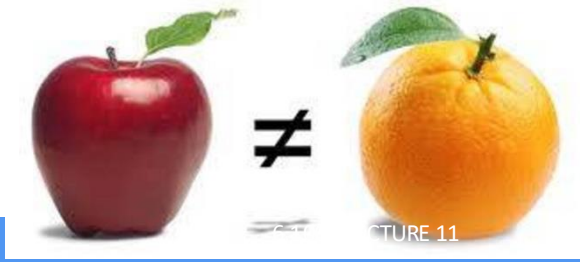
Spacing on y-axis doesn't match numbers

Same data

Moral: Look carefully at the axes labels and scales

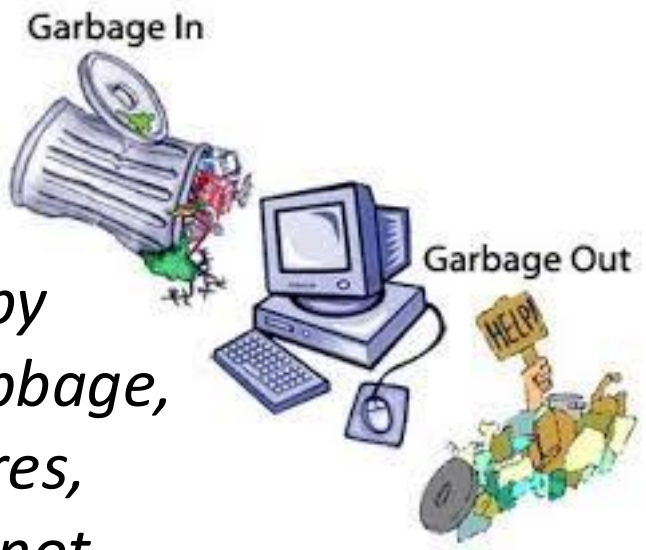
One Picture: Multiple Deceptions

Moral:

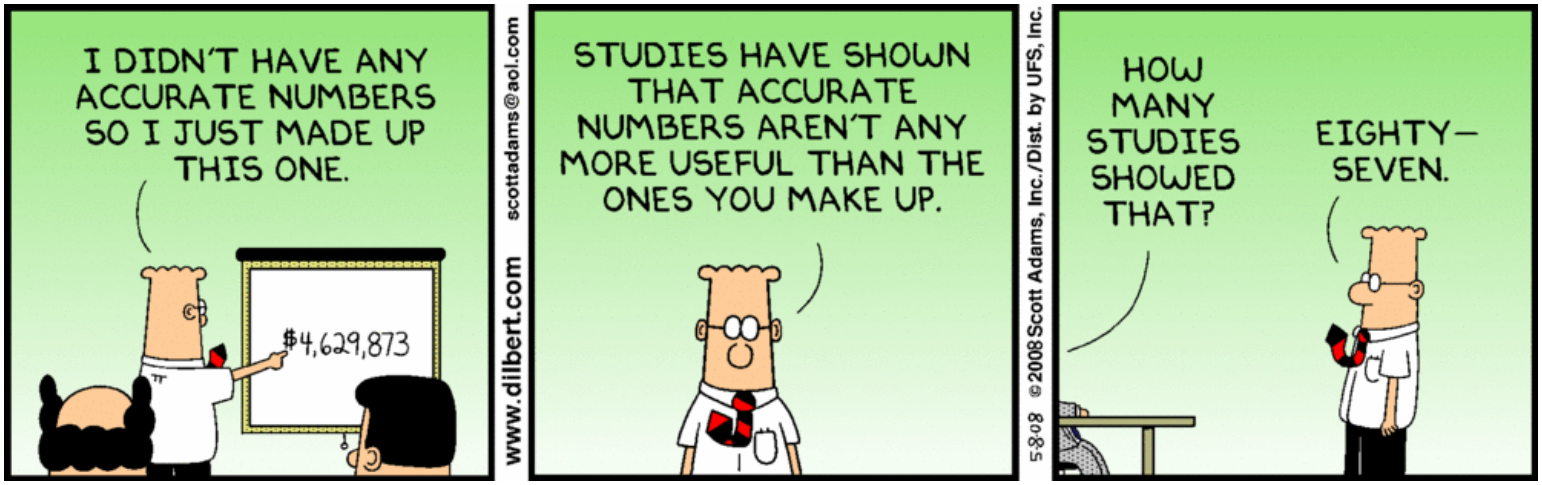


Leaving Plotting

Garbage In, Garbage Out



"On two occasions I have been asked [by members of Parliament], 'Pray, Mr. Babbage, if you put into the machine wrong figures, will the right answers come out?' I am not able rightly to apprehend the kind of confusion of ideas that could provoke such a question." – Charles Babbage (1791-1871)



Data is Often Wrong

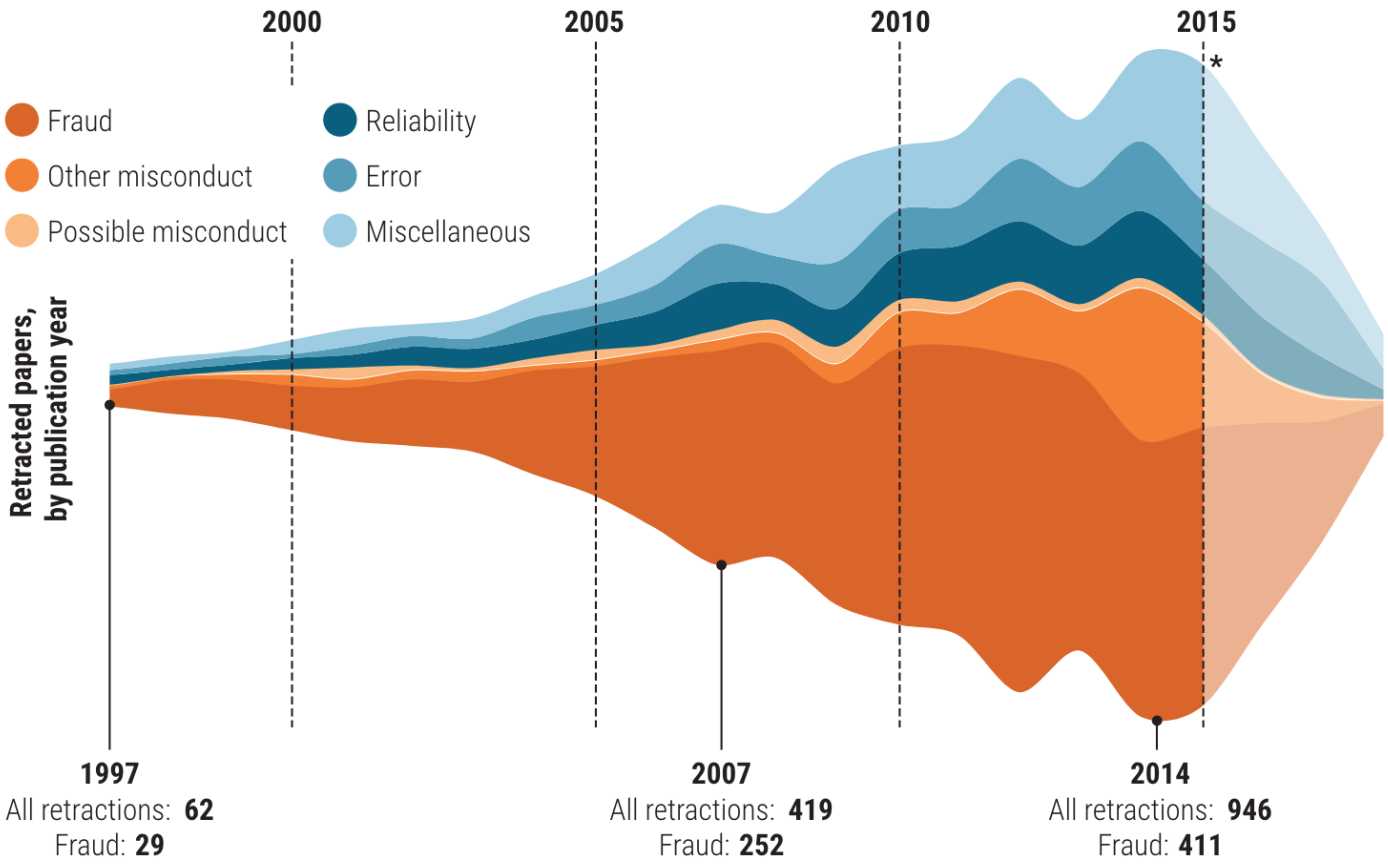
- Poorly measured
  - Bad questionnaires
  - Bad instrumentation
  - Human error
- Inaccurately recorded
  - Transcription errors
- Outdated
- Incomplete
- Generated by flawed software
- Intentionally falsified
- …

The Stanford Daily

News • University

Internal review found 'falsified data' in Stanford President's Alzheimer's research, colleagues allege

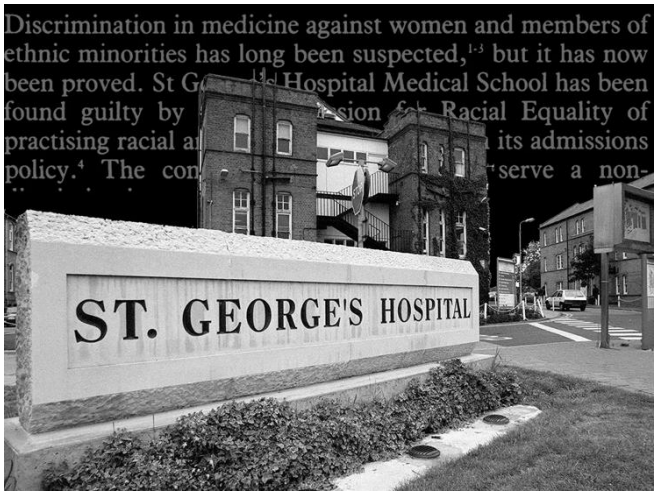
And It's Not Just at Stanford



https://www.science.org/content/article/what-massive-database-retracted-papers-reveals-about-science-publishing-s-death-penalty

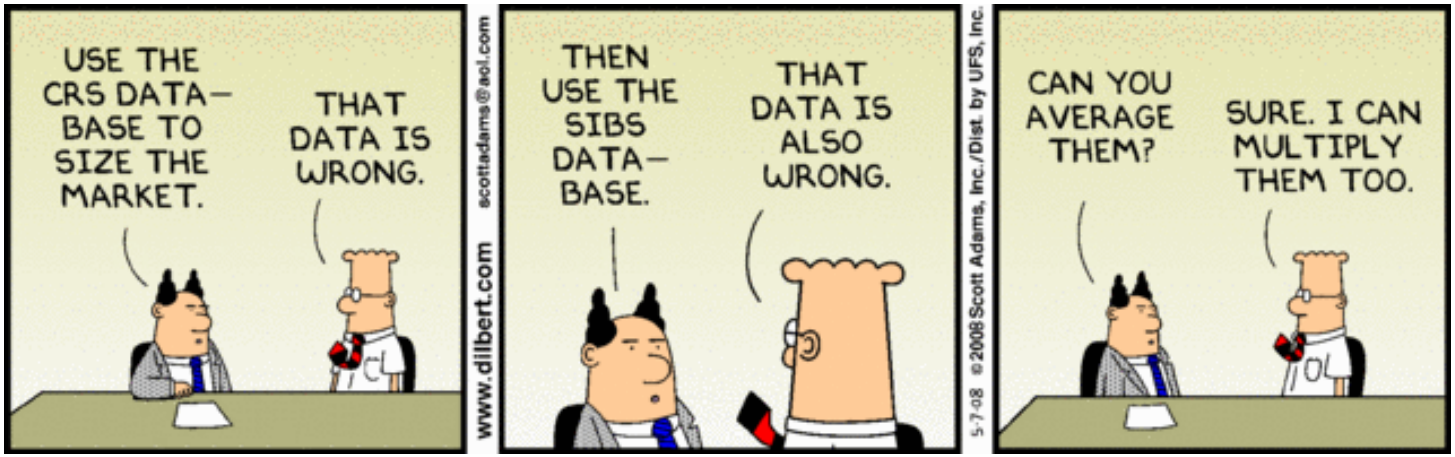
Even Accurate Data Can Be Garbage

- System built to score applicants to a medical school in UK
- Learned from multiple years of admissions data
- 90-95% agreement when tested on holdout set
- Codified patterns in previous data
  - Non-European name: - 15 points
  - Female: - 3 points

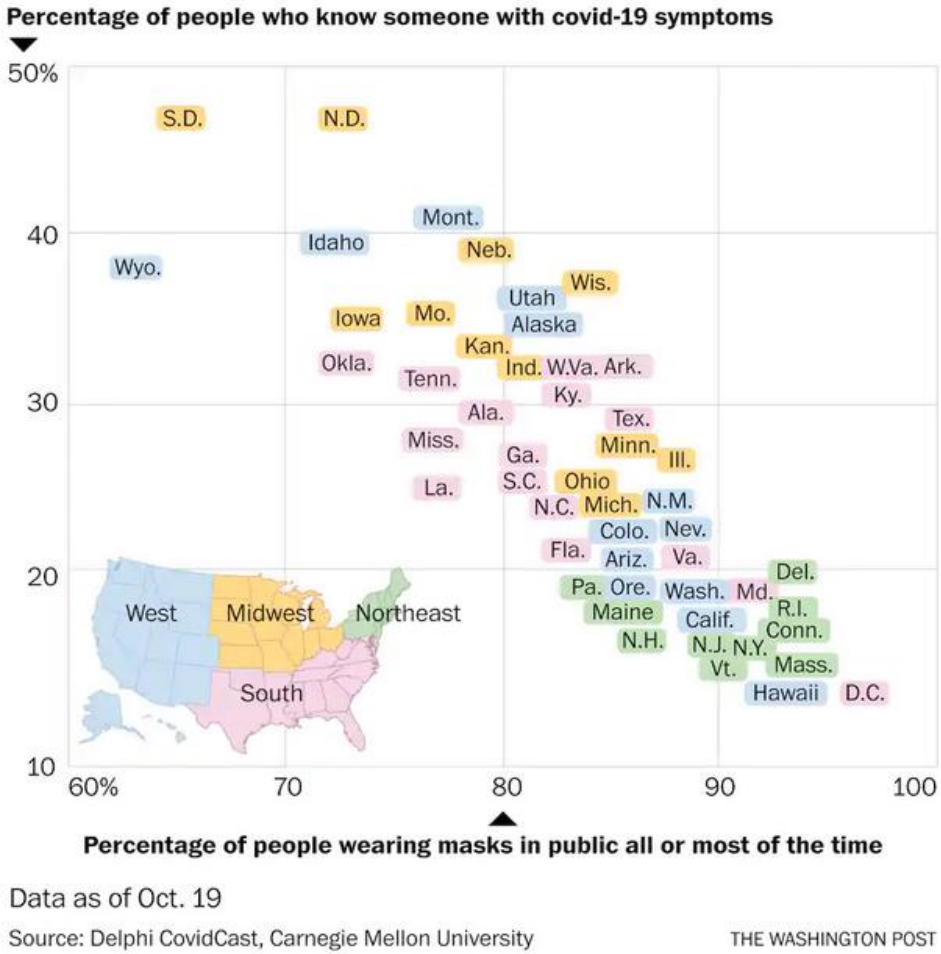


Poor Quality Data Not Always Useless

- Many useful techniques for cleaning data
- But there are problems that cannot be overcome
  - Lack of signal
  - Systematic corruption
- Imagine using grades in 6.100A to assess whether men or women are better at programming
  - Grades assigned at random?
  - Grades corrupted by randomly raising or lowering earned grade?



Cum Hoc, Ergo Propter Hoc



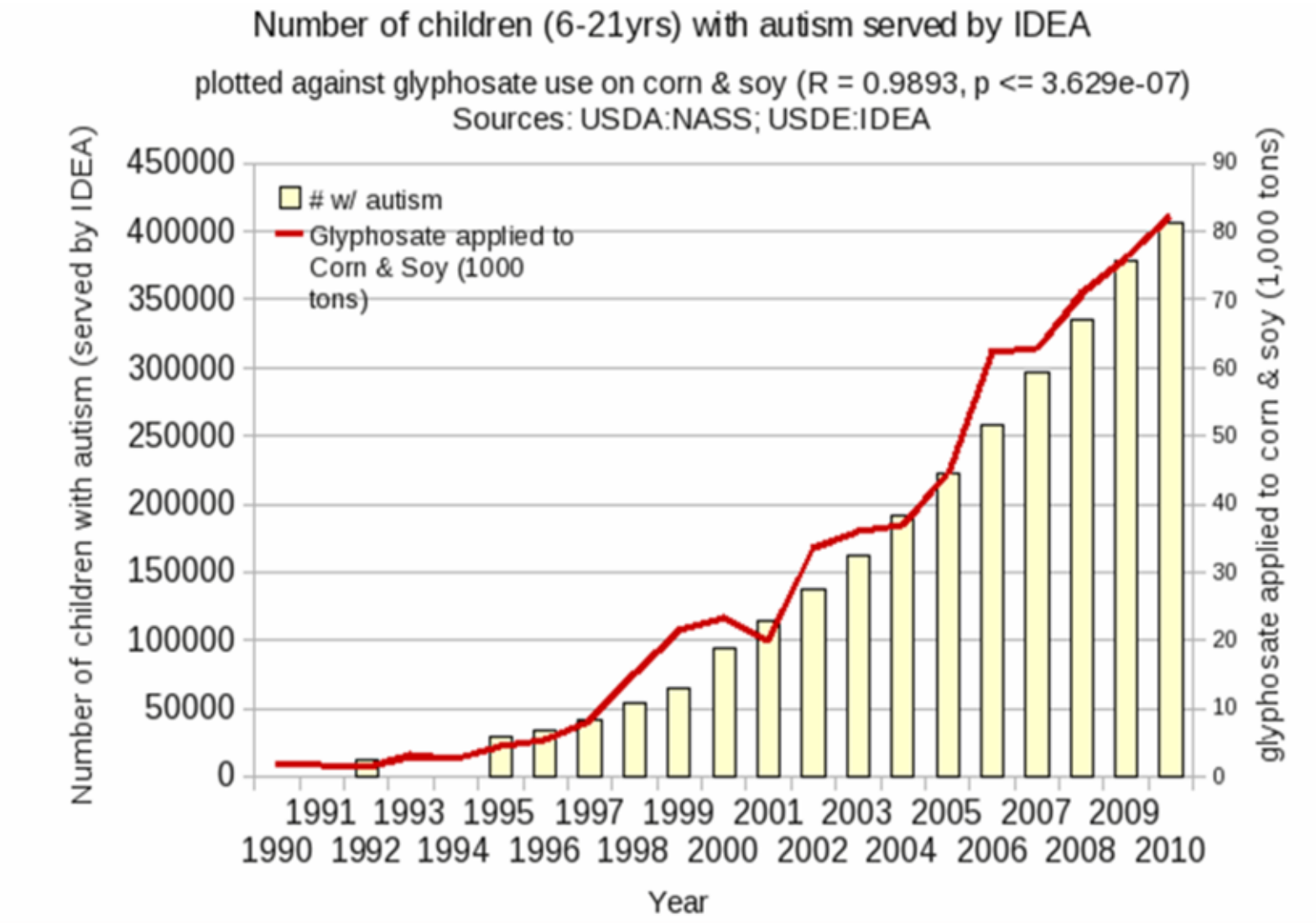
Use linear regression to fit line, R**2 = 0.73

Appears that knowing someone with Covid-19 leads to not wearing a mask

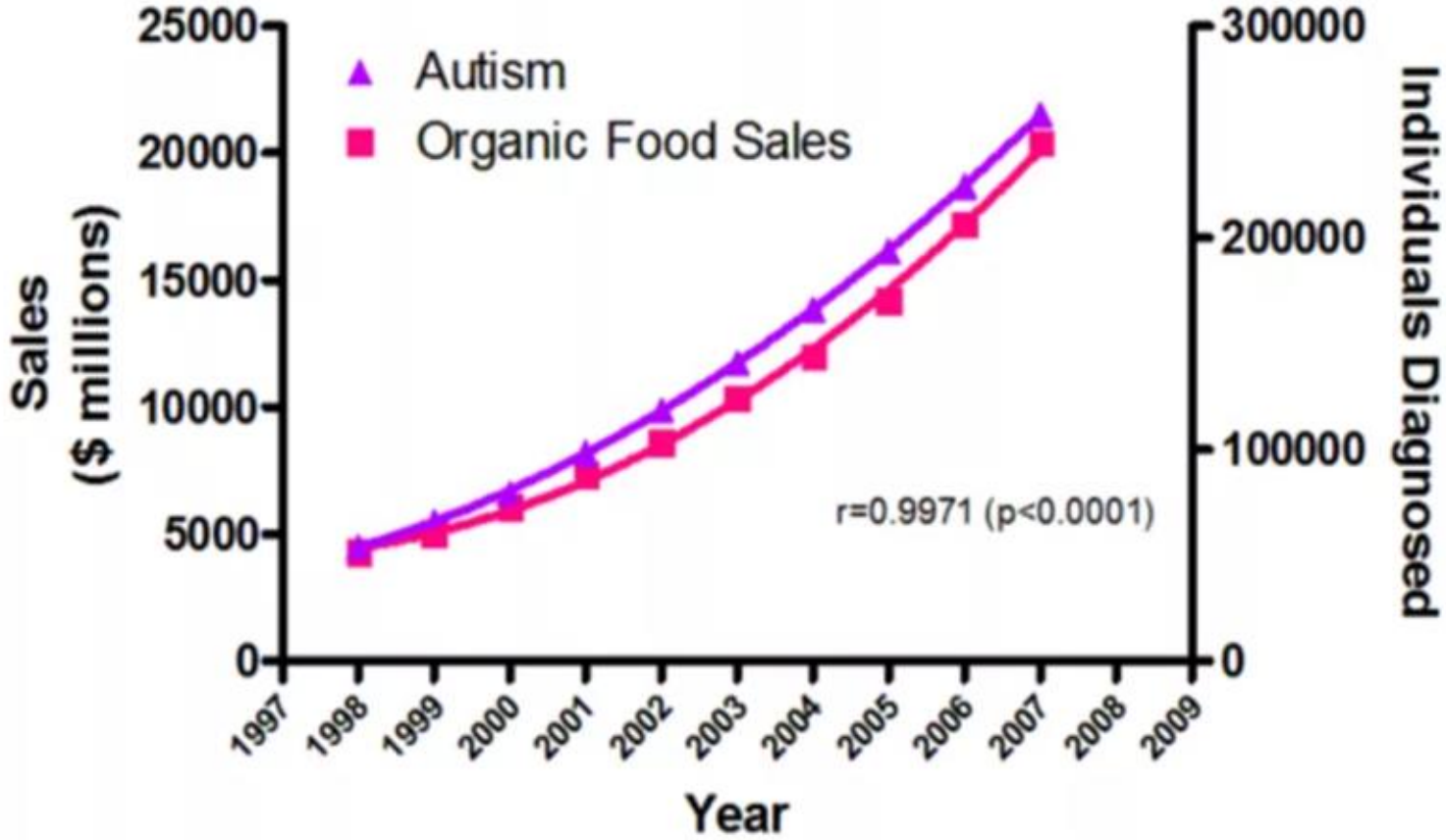
Or maybe not wearing a mask leads to knowing someone with Covid

Moral: Correlation does not imply causation!

A Cause of Autism?



A Cause of Autism?

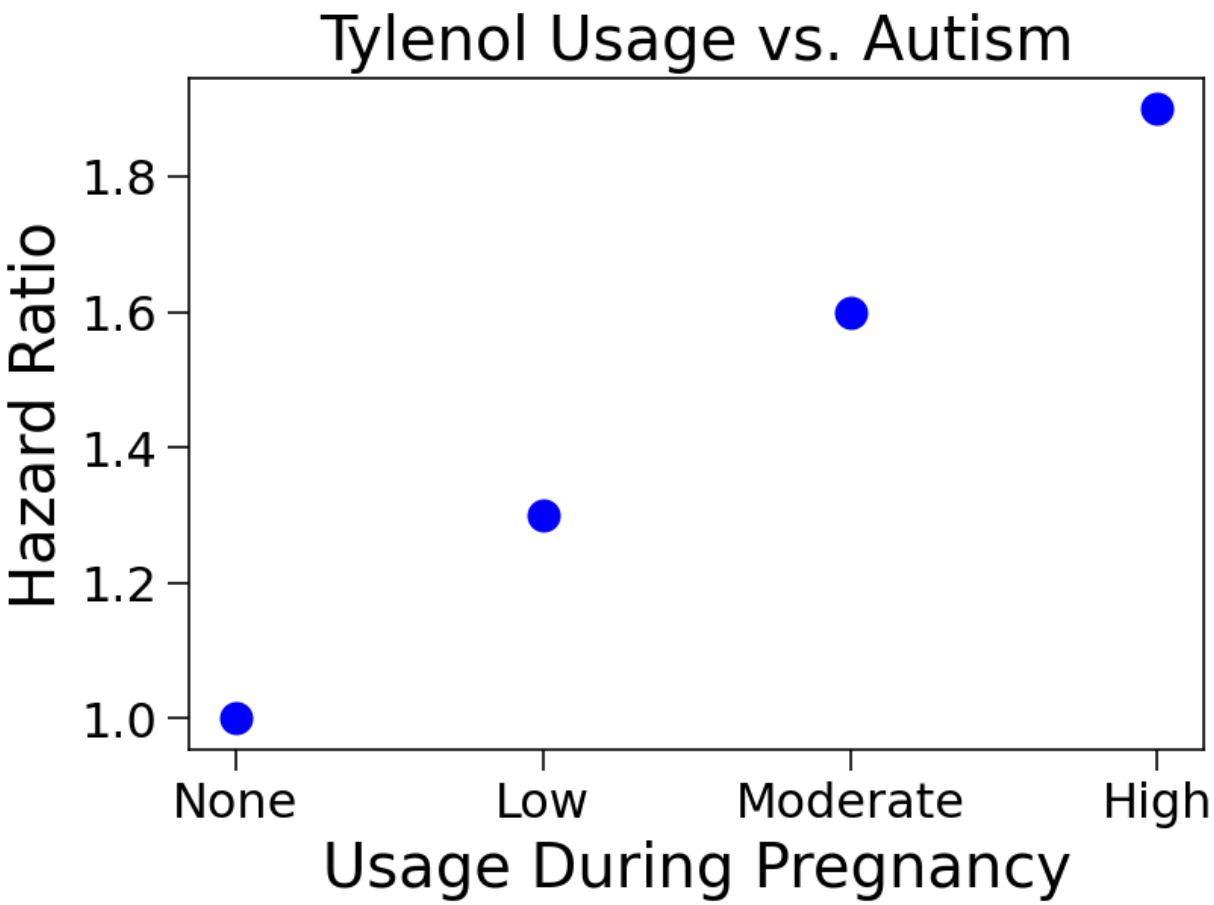


Sources: Organic Trade Association, 2011 Organic Industry Survey; U.S. Department of Education, Office of Special Education Programs, Data Analysis System (DANS), OMB# 1820-0043: "Children with Disabilities Receiving Special Education Under Part B of the Individuals with Disabilities Education Act

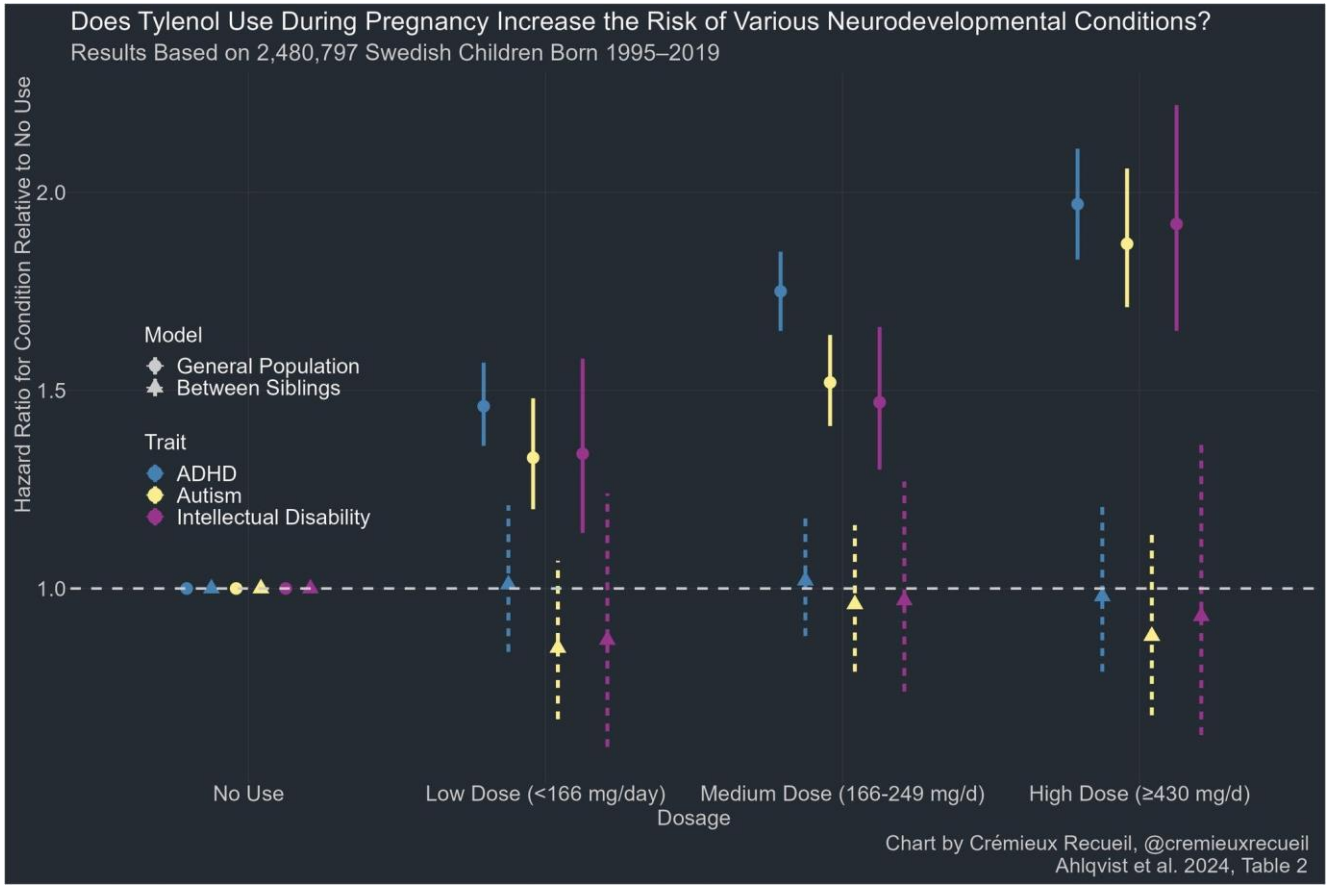
Tylenol Use During Pregnancy and Autism

September 2025

A Large Study Showed a Strong Correlation



Study Providing Data for Previous Chart



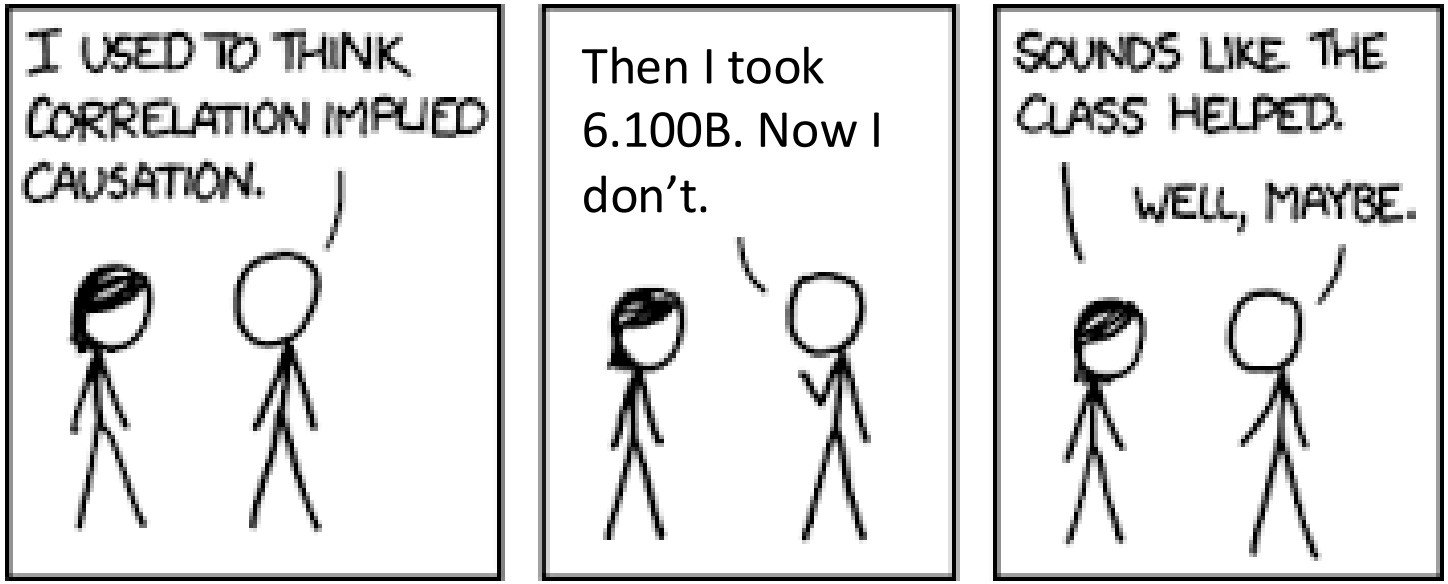
Confounders matter, especially when using observational data, e.g.,

Were mothers who took more Tylenol sicker?

Were they more prone to taking other drugs?

This study used siblings as a control

Randomized control trials the gold standard


I USED TO THINK CORRELATION IMPLIED CAUSATION.
Then I took 6.100B. Now I don't.
SOUNDS LIKE THE CLASS HELPED.
WELL, MAYBE.